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Notice of the General Office of the People's Government of Hunan Province on Issuing the "Hunan Province's 14th Five-Year Plan for Ecological and Environmental Protection"

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To the people's governments of all cities, prefectures, counties, districts, and municipalities, and all departments and committees of the provincial government and its directly affiliated agencies:

The "Hunan Province 14th Five-Year Plan for Ecological and Environmental Protection" has been approved by the Provincial People's Government and is now issued to you. Please organize and implement it conscientiously.

General Office of the People's Government of Hunan Province

September 30, 2021

(This document is disclosed voluntarily)

Hunan Province's "Fourteenth Five-Year" Ecological and Environmental Protection Plan

The 14th Five-Year Plan period is the first five years of embarking on a new journey of building a modern socialist country in an all-round way and marching towards the second centenary goal. It is also a critical five years for our province to vigorously implement the "three highs and four new" strategy and strive to build a modern new Hunan. Ecological and environmental protection is the key to the overall promotion of the "five

It is an important part of the overall layout of "integration", an important means to promote high-quality development, and an important factor in meeting the people's growing needs for a better life. In order to thoroughly implement the spirit of General Secretary Xi Jinping's important speech during his inspection of Hunan, continuously improve the ecological environment quality of the province, and build a harmonious coexistence between man and nature,

This plan is formulated in accordance with the overall deployment of the National "14th Five-Year Plan for Ecological and Environmental Protection" and the "14th Five-Year Plan for National Economic and Social Development of Hunan Province and the Outline of the Vision for 2035".

1. The basis of the 14th Five-Year Plan

During the 13th Five-Year Plan period, the provincial party committee and government adhered to the guidance of Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era, thoroughly implemented Xi Jinping Thought on Ecological Civilization, resolutely implemented the decisions and arrangements of the Party Central Committee and the State Council, grasped the new development stage, implemented the new development concept, and built a new development pattern.

We have promoted high-quality development, solidly advanced the three major battles for blue sky, clear water and clean land and seven major landmark campaigns, actively promoted the rectification of prominent ecological and environmental problems, deepened the reform of the ecological civilization system, and made every effort to serve high-quality development. The ecological environment quality of the province has been significantly improved, and environmental protection has benefited the people.

The degree of environmental protection has continued to improve, the environmental governance capacity has continued to increase, and the level of green development has continued to improve, which has laid a solid green foundation and quality for building a well-off society in an all-round way.

1. Current Development Status

The concept of ecological protection priority has become more prominent. The provincial party committee and the provincial government have resolutely shouldered the political responsibility of ecological environmental protection and established the strategy of building an ecologically strong province. The provincial party committee issued the "Decision on Adhering to Ecological Priority and Green Development, Deeply Implementing the Development Strategy of the Yangtze River Economic Belt, and Vigorously Promoting the High-Quality Development of Hunan".

The Provincial Party Committee and the Provincial Government issued the "Implementation Opinions on Comprehensively Strengthening Ecological and Environmental Protection and Resolutely Winning the Tough Battle Against Pollution", formulated the "Three-Year Action Plan for the Tough Battle Against Pollution in Hunan Province", vigorously promoted the tough battle against pollution, carried out the "Summer Offensive" for five consecutive years, and implemented the Xiangjiang River Protection Plan.

The three "three-year action plans" for the protection and governance of Dongting Lake, the special governance of the ecological environment of Dongting Lake, the construction of urban and rural environmental infrastructure, and the governance of the Xiangjiang River Basin and Dongting Lake mountain, forest, field, lake and grass system. The provincial ecological environment protection inspection, the assessment of the effectiveness of the pollution prevention and control battle, and the natural

In the outgoing audit of resources and assets, the leading Party and government cadres at all levels in the province have firmly established the concept of ecological priority and green development. The intensity of pollution control, the practicality of work measures and the significant changes in the environment are unprecedented.

The ecological and environmental protection system has been further improved. The provincial party committee and the provincial government have established the provincial leading group for rectification of outstanding ecological and environmental problems, the provincial ecological and environmental protection committee, and the provincial ecological and environmental protection supervision leading group to comprehensively strengthen overall coordination.

The Hunan Provincial Department of Ecology and Environment has enacted the Regulations on the Responsibility of the Party and Government for Ecological and Environmental Protection, the Measures for Accountability for Major Ecological and Environmental Issues (Incidents) in Hunan Province, and the Implementation Measures for Ecological and Environmental Protection Supervision in Hunan Province, etc., to build a responsibility system of "Party and government share the same responsibility, one post with two responsibilities" and a pattern of joint management of ecological and environmental protection.

Civilized system reform, basic construction of a number of systems such as natural resource asset property rights, "three lines and one list", pollution discharge permits, ecological compensation, ecological environment damage compensation, river (lake) chief system, etc., comprehensive completion of the vertical management system reform of environmental protection agencies below the provincial level for monitoring, supervision and law enforcement, and promotion of the "four strict and four basic" three-year

Action plan, the basic capacity for ecological and environmental protection has been strengthened.

The improvement in ecological environment quality is more obvious. In 2020, the water quality rate of 60 national assessment sections in the province was 93.3%, and the national assessment sections completely eliminated the inferior Class V water quality. The Hunan section of the Yangtze River mainstream and the Xiang-Zi-Yuan-Li mainstream all reached or exceeded Class II water quality, and the total phosphorus concentration in Dongting Lake dropped to 0.06 millimolar.

The proportion of days with good air quality in prefecture-level cities in the province was 91.7%, an increase of 10.3 percentage points over 2015. The annual average concentration of PM2.5 in the province was 35 micrograms per cubic meter. The number of cities that met the national ambient air quality level II standard increased from 0 in 2015 to 1.3 in 2016.

From 2010 to 2017, the safe utilization rate of polluted cultivated land and contaminated plots in the province has reached the national assessment requirements, a detailed investigation of the soil pollution status of agricultural land in the province has been completed, 428 key industries involving cadmium and heavy metals have been investigated and rectified, and the soil environment is safe and controllable.

Ecological protection and restoration efforts have become more solid. We have vigorously promoted the rectification of "Green Shield" issues, launched the National Greening Action, comprehensively promoted the construction of ecological corridors, strengthened the restoration of ecologically fragile areas, strengthened biodiversity protection, and promoted the ban on fishing and withdrawal of fishing in key waters of the Yangtze River Basin. By the end of 2020, the province's forest cover will be

The coverage rate is greater than 59%, the forest volume is 588 million cubic meters, an increase of 108 million cubic meters over the end of 2015, the wetland protection rate is 75.77%, an increase of 3.27% over the end of 2015, and a total of 386,400 mu of European and American poplars have been cleared in the Dongting Lake area, and 444,200 mu of land, beaches and coastlines have been repaired and cleared;

255 tailings ponds and 545 abandoned mines in the Yangtze River Economic Belt have been built, 9,795 hectares of mine ecological restoration has been completed, and 55 national-level green mines have been built; the ecological environment of 54 national key ecological function zones has remained generally stable, and rare species such as black storks, Chinese mergansers, and white-tailed sea-eagles have been

The populations of rare and endangered animals and rare and endangered plants such as the silver fir, southern redwood, and bole tree remained stable; the numbers of endemic rare and endangered species such as the elk, finless porpoise, little swan, and white spoonbill remained stable; and the number of migratory birds wintering in Dongting Lake increased significantly.

The outstanding ecological and environmental problems have been solved more thoroughly. Taking the rectification of problems reported by the central environmental protection inspection and the "look back", the rectification of problems disclosed in the ecological and environmental warning film of the Yangtze River Economic Belt, and the rectification of problems reported by provincial environmental protection inspection as an opportunity, we will strengthen responsibility and accelerate the resolution of outstanding ecological and environmental problems left over from history.

The low enclosure of Xiasai Lake in Dongting Lake area, small hydropower in Zhangjiajie Giant Salamander National Nature Reserve, illegal construction of Changsha-Zhuzhou-Tanzhou "Green Heart", ecological damage in Hengyang Dayishan Nature Reserve, environmental pollution in Yiyang coal mines, overdue storage of hazardous solid waste, pollution in tailings ponds and tailings ponds, and other stubborn problems have been exposed.

to effective control.

The role of promoting high-quality development has been more fully exerted. The supply-side structural reform has been deepened, and the guiding, optimizing, forcing and promoting role of ecological and environmental protection has been significantly enhanced. Environmental standards have been used to continuously eliminate obsolete production capacity, and five key areas in the Xiangjiang River Basin, such as Qingshuitang in Zhuzhou and Xikuangshan in Loudi, have achieved green development.

The province has actively promoted the relocation and upgrading of the chemical industry along the Yangtze River and the rectification of "scattered, messy and polluting" enterprises. The total output value of the province's environmental protection industry is nearly 300 billion yuan. Provincial and above parks (including provincial industrial concentration areas) have gathered nearly 70% of the province's scale industrial added value, and the industrial structure and layout have been further optimized.

In 2020, the utilization rate of fertilizers and pesticides in the province will reach 40%, the comprehensive utilization rate of straw will reach 80%, the resource utilization rate of breeding waste will reach 75%, and the recycling rate of agricultural film will reach 81%, and the level of green agricultural development will be further improved. During the "Thirteenth Five-Year Plan" period, the total amount of pollution emissions continued to decrease, and the province

The province's chemical oxygen demand, ammonia nitrogen, sulfur dioxide, and nitrogen oxide emissions decreased by 11.6%, 17.1%, 28.7%, and 18.8% respectively compared with 2015. Carbon dioxide emissions per 10,000 yuan of GDP decreased by 19.8%, energy consumption per unit of regional GDP decreased by 19.4%, and the added value of industrial units increased by 1.3%.

The cumulative energy consumption has decreased by 27.47%.

The people have a stronger sense of gain. We were the first in the country to complete the environmental remediation of drinking water sources at the county level and above, and we have fully completed the demarcation of the "1,000 tons per 10,000 people" centralized drinking water source protection areas in towns and villages. We have completed the comprehensive rural environmental governance of 13,000 administrative villages, and the harmless treatment of urban domestic waste.

The treatment rate reached 99.58%. The proportion of black and odorous water bodies eliminated in the built-up areas of prefecture-level cities in the province reached 98.4%. 11 cities and counties were rated as national ecological and civilization demonstration counties, 26 counties and districts were rated as provincial ecological and civilization demonstration counties, and 2 counties and cities won the national "Green Water and Green Mountains are Gold and Silver Mountains" practice innovation

The province has added 7 national garden cities (county towns), Yueyang City has been approved as a green development demonstration zone of the Yangtze River Economic Belt, Chenzhou City has been approved as a national sustainable development agenda innovation demonstration zone, Changde City has been established as a national sponge city and an international wetland city, and Loudi and other 5 cities have been awarded the title of "National Greening Demonstration Zone".

Nine cities including Yiyang were awarded the title of "National Forest City". Blue sky and white clouds have become the norm, and green waters and green mountains can be seen everywhere.

Table 1 Completion of the main indicators of the 13th Five-Year Plan

Indicator Classification	Indicator name	2015	2020 Planning objectives	2020 Achievement of planning goals
Air quality quantity	Number of days with good air quality in prefecture-level cities (including Jishou) Proportion(%)	81.4	82.9	91.7
	The proportion of days with good air quality in counties (cities) (%) The annual	–	85	96.3
	average concentration of PM2.5 in urban areas of prefecture-level cities (including Jishou) Degree (μg/m3)	51	44ÿ-18%ÿ	35
Water environment quality	Yangtze River system (national controlled sections) Water quality rate ÿ%ÿ	81.3	93.2	93.2
	Pearl River system (national controlled sections) (%)	–	100	100
	Dongting Lake (national controlled section) (%)	–	Total phosphorus type IV, the rest refers to Class III	Total phosphorus type IV, the rest refers to Better than Class III
	Centralized drinking water sources in prefecture-level cities (including Jishou) Ground water quality compliance rate (%)	96.7 Meets national assessment requirements		100 (Note: 2020, Except for the sudden Sudden environmental events affect water In addition to water sources that meet the standards, other water sources All met the criteria.)
Ecological environment condition	Forest cover rate(%)	59.57	Maintain above 59.0%	ÿ59.0%
Main pollutants Emission reduction targets	Chemical oxygen demand	–	10.1	11.6
	(%) Ammonia	–	10.1	17.1
	nitrogen (%) Sulfur	–	21	28.7
	dioxide (%) Nitrogen	–	15	18.8
–	oxides (%) Reduction in carbon dioxide emissions per unit of regional GDP ÿ%ÿ	–	18	19.8

During the 13th Five-Year Plan period, the province's ecological and environmental protection work has achieved remarkable results. However, overall, the structural, root and trend pressures on ecological and environmental protection in our province have not been fundamentally alleviated, the severe situation of pollution emissions and ecological damage has not fundamentally changed, and ecological and environmental incidents have occurred frequently.

The high-risk situation of the outbreak has not fundamentally changed, the modern governance system is still imperfect, and the turning point from quantitative change to qualitative change in the improvement of ecological environment quality has not yet arrived. Compared with the overall requirements of the province's high-quality economic and social development and the construction of a modern new Hunan where people and nature coexist in harmony, ecological and environmental protection still has a long way to go.

There are certain gaps, mainly in:

The improvement of environmental quality is not stable. The overall water environment quality has improved significantly, but some sections of the Xiang-Zi-Yuan-Li River still have water quality exceeding the standard or declining. The decline in the total phosphorus concentration in Dongting Lake has slowed down, and it is difficult to reach the Class III standard in the short term. The hidden danger of excessive heavy metals such as antimony and thallium in drinking water sources is still high.

The improvement of ambient air quality is unstable, PM2.5 concentration is still high, ozone concentration is rising instead of falling, heavy pollution weather occurs from time to time, and the overall urban air quality has not yet gotten rid of the characteristics of "meteorological influence". The soil environment safety situation is severe, the area of polluted arable land is large, and key industries are

The land use rate of enterprises exceeds the standard, and there are many and extensive waste residue sites left over from history. The pollution load from agricultural production and rural life is still heavy, and the black and odorous water bodies in cities have not yet been cleaned for a long time. There is still a gap between the improvement of ecological environment quality and the people's needs for a better life.

The level of green development needs to be improved. Overall, the province's industrial structure, which is dominated by nonferrous metals, smelting, chemical industry, and cement building materials, has not changed fundamentally, and the added value of the six high-energy-consuming industries still accounts for a high proportion of scale industries; the energy structure dominated by coal has not changed fundamentally, and coal consumption accounts for 1.3% of the total energy consumption.

Reached 55%; transportation is heavily dependent on roads, and the province's road freight volume accounts for more than 80% of the total social transportation volume; agricultural development intensity is high, the combination of planting and breeding is insufficient, and the efficiency of waste resource utilization is low; green consumption, low-carbon travel, garbage classification and other green lifestyles have not yet been effectively formed. Implementation

The task of implementing the green development concept, optimizing the industrial structure, energy structure, transportation structure, and land use structure, and promoting the green and low-carbon transformation of production and life is arduous.

The task of addressing historical environmental problems is arduous. During the 13th Five-Year Plan period, although a number of prominent ecological and environmental problems were solved, there are still many historical issues in ecological and environmental protection in our province, including heavy heavy metal pollution in the soil, many problems left over from industrial mining pollution, prominent non-point pollution in agriculture and rural areas, and the Dongting Lake area.

The quality of the ecological environment needs to be improved urgently, the construction of urban and rural environmental infrastructure is lagging behind, and the basic capacity for ecological environment governance is weak. At the same time, environmental risks such as polluted land, hazardous wastes, abandoned mines and mine water pollution, and natural ecological damage still exist, and the task of ecological environment safety management is arduous.

huge.

The basic capacity for ecological and environmental protection is still relatively weak. The responsibilities of "Party and government share the same responsibility, one post has two responsibilities" and "three controls and three musts" for ecological and environmental protection need to be further tightened and consolidated, and the internal motivation of relevant responsible entities to perform their duties has not been effectively stimulated; the reform of the vertical management system of the ecological environment has only undergone "physical changes".

The reform dividend has not been fully released. The mechanism for realizing the value of ecological products that green mountains and clear waters are gold and silver mountains has not yet been formed. The green price system and the institutional systems such as pollution discharge rights and carbon emission rights need to be further established and improved. The investment in ecological and environmental governance is insufficient, the channels are single, and the market mechanism is not yet

The ecological environment monitoring capacity is not strong, the level of informatization is not high, the comprehensive law enforcement force has not yet been reflected, the support of science and technology needs to be strengthened, and the awareness of ecological and environmental protection actions of the whole people needs to be improved.

(2) Situation requirements.

During his inspection tour of Hunan, General Secretary Xi Jinping made important instructions on building "three highlands" and practicing the "four new" missions, which anchored new goals and missions for the province's ecological and civilized construction, provided a fundamental guideline for building a modern new Hunan where people and nature coexist harmoniously, injected strong impetus, and promoted the development of Hunan.

The Central Committee of the Communist Party of China requires the complete, accurate and comprehensive implementation of the new development concept, making major strategic decisions on carbon peak and carbon neutrality, promoting high-quality development of the Yangtze River Economic Belt, and building a beautiful central region with green development, etc., which will bring great benefits to the province's ecological environment.

The new environment protection work has brought new opportunities and challenges; the people's eager expectation for building a modern new Hunan where people and nature coexist harmoniously has put forward new tasks and new requirements for ecological and environmental protection work. Overall, the province's ecological civilization construction has entered a strategic direction of focusing on carbon reduction and promoting

It is a critical period for synergizing pollution reduction and carbon reduction, promoting the overall green transformation of economic and social development, and achieving a qualitative change in the improvement of ecological environment quality. During the "14th Five-Year Plan" period, we must resolutely implement Xi Jinping's thoughts on ecological civilization and the spirit of General Secretary Xi Jinping's important speech during his inspection of Hunan, vigorously implement the "three highs and four

We will implement the "new" strategy, comprehensively strengthen ecological civilization construction, fight hard to prevent and control pollution, promote the modernization of the ecological environment governance system and governance capacity, seize opportunities, work hard, innovate, and show new achievements in building a modern new Hunan where people and nature live in harmony.

II. Overall requirements for the 14th Five-Year Plan

1. Guiding ideology

We will continue to be guided by Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era, fully implement the spirit of the 19th CPC National Congress and the Second, Third, Fourth, and Fifth Plenary Sessions of the 19th CPC Central Committee, thoroughly implement Xi Jinping Thought on Ecological Civilization and the important speech delivered by General Secretary Xi Jinping during his inspection tour of Hunan, and grasp the new development We will implement the new development concept, build a new development pattern, closely focus on the strategic positioning of "One Belt and One Road" and the strategic goals of "Three Highs and Four New", adhere to the people-centered approach, take the comprehensive green transformation of economic and social development as the guide, take serving the high-quality development of the economy as the main line, and continuously improve the ecological environment. With quality as the core and pollution reduction and carbon reduction as the overall grasp, we will focus on promoting green and low-carbon development, fully promote carbon peak action, fight hard to prevent and control pollution, prevent and resolve ecological and environmental risks, enhance the functions of the ecological system, promote the modernization of the ecological and environmental governance system and governance capacity, and promote ecological Civilization construction has reached a new level, demonstrating new achievements in building a modern new Hunan where people and nature coexist in harmony.

(2) Basic principles

Adhere to the leadership of the Party. Comprehensively strengthen the Party's leadership over ecological and environmental protection work, implement the principles of "Party committee leadership, government guidance, enterprise main body, public participation", and "Party and government common responsibility", "one post, two responsibilities", "three controls and three musts", urge enterprises to fulfill their main responsibility for pollution control, guide and mobilize the masses to participate in the work, and ensure that the Party and government have the same responsibility, and .. Universal participation.

Adhere to the principle of ecology for the people. Continue to improve the quality of the ecological environment, adhere to the organic combination of ecological priority, high-quality development and creating a high-quality life, accelerate the resolution of prominent ecological and environmental problems around the people, create a good production and living environment for the people, and provide more high-quality ecological products. products, and continuously meet the people's growing needs for a beautiful ecological environment.

Adhere to green development. Unswervingly implement the new development concept, maintain the strategic focus of ecological civilization construction, correctly grasp the relationship between protecting the ecological environment and developing the economy, promote structural adjustment, optimize the national land space development pattern, promote carbon peak and carbon neutrality, and adhere to the Yangtze River Economic Belt "together grasp the great The general rule is to protect the environment and not engage in large-scale development.

Adhere to systematic governance. Strengthen the protection and restoration of mountains, rivers, forests, fields, lakes, grasslands and deserts, improve the overall and holistic nature of governance measures, increase structural adjustment efforts, coordinate efforts to address climate change and improve ecological and environmental quality, coordinate control of fine particulate matter and ozone, coordinate ecological protection and pollution prevention, and strengthen Comprehensive management of environmental elements.

Adhere to the bottom line thinking. Actively do a good job in ecological and environmental risk prevention and mitigation, integrate risk prevention into the entire process of ecological and environmental protection, strengthen source prevention and control, properly handle sudden environmental incidents, firmly hold the bottom line of ecological and environmental safety, and effectively protect the lives and health of the people. healthy.

Adhere to reform and innovation. Promote the modernization of the ecological environment governance system and governance capacity, strengthen the construction of ecological environment rule of law, continue to deepen the reform of the ecological environment system and mechanism, improve the market-oriented mechanism of ecological environment governance, make up for the shortcomings of the ecological environment governance system and mechanism, and form a clear-headed and An ecological and environmental governance system with scientific decision-making, strong execution, effective incentives, diverse participation, and benign interaction.

(III) Main objectives

Looking ahead to 2035, the environmental quality will be fundamentally improved, the ecological barrier will be more solid, green production and living methods will be widely formed, prominent ecological and environmental problems will be basically solved, carbon peak will be achieved and carbon neutrality will be promoted, an ecologically strong province will be basically built, the goal of building a beautiful Hunan will be basically achieved, and people and nature will live in harmony. It will basically meet the people's needs for a beautiful ecological environment in their pursuit of a high-quality life, and basically realize the modernization of the ecological environment governance system and governance capabilities.

The overall goal of the 14th Five-Year Plan. The green transformation of production and lifestyle has achieved remarkable results, the allocation of energy resources has become more reasonable, the utilization efficiency has been greatly improved, the national spatial development and protection pattern has been optimized, the total amount of pollutant emissions has continued to decrease, and the quality of the ecological environment has continued to improve. Improve and highlight ecological environment issues The resolution of major ecological and environmental risks has been accelerated, the ecological safety barrier has been basically resolved, the urban and rural living environment has been significantly improved, the ecological environment governance system and governance capacity have been significantly enhanced, and new progress has been made in the construction of ecological civilization.

Specific indicators of the "14th Five-Year Plan": By 2025, the province's green and low-carbon development level will be significantly improved, the total emission of key pollutants, carbon dioxide emissions per unit of regional GDP, and energy consumption will continue to decrease; the quality of the water environment will continue to improve, and comprehensive elimination of The total phosphorus concentration in Dongting Lake remains inferior to Class V water. The water quality of centralized drinking water sources in cities at the municipal level has fully met the standards, and the compliance rate of centralized drinking water sources in county-level cities has reached 95.8%. The air quality has continued to improve, and the annual average concentration of PM2.5 in cities at the prefecture level and above has continued to decline, and severe and above pollution days have been basically eliminated. number; the province's soil pollution environmental risks have been effectively controlled, the level of safe soil utilization has been consolidated and improved, polluted cultivated land and key construction land have been safely utilized and effectively controlled; natural ecological protection supervision has made positive progress, and the forest coverage rate has not decreased.

Table 2 Indicator system for the "14th Five-Year Plan" ecological and environmental protection plan

sequence Number	index Classification	Indicator name		Unit 2020		2025	Attributes
1	green Low Carbon	Reduction in carbon dioxide emissions per unit of GDP		%	–	The state issues binding indicators	
2		Reduction in energy consumption per unit of		%	–	The state issues binding indicators	
3		GDP Ratio of non-fossil energy to primary energy		%	21	23	Anticipation
4	environment quality	Surface water	consumption % of surface water reaching or better than Class III %		93.3 The state issues binding indicators		
5			of surface water inferior to Class V % Dongting Lake (national		0	0	Constraints
6			controlled section, total phosphorus)	mg/l	0.06 The expected performance of the national indicators		
7		Groundwater Ratio of groundwater quality Class V water		%	5	5	Anticipation
8		centralized drinking water source	Centralized drinking water sources in prefecture-level cities Quality compliance	%	100	100	Anticipation
9			rate of centralized drinking water sources in county-level cities Quality compliance	%	91.7	95.8	Anticipation
10		Atmospheric environment territory	rate Annual average PM2.5 concentration in cities at prefecture level and above Degree decrease	%	–	The state issues binding indicators	
11			The annual average concentration of PM2.5 is less than 35μg/m3 Number of city-level cities	indivual	7	10	Anticipation
12			Days with good air quality in cities at prefecture level and above Ecological	%	91.7 The state issues binding indicators		
13 Ecology		Quality Index (New EI) Forest Coverage		–	–	Stable and positive expectations	
14	Function			%	ÿ59	ÿ59	Constraints

15		Wetland protection rate		%	75.77 ÿ	The anticipation of national indicators	
16		The proportion of ecological protection red line to land area		%	20.23 ÿ	No reduction	Anticipation
17	risk Prevention and Control	Safe utilization rate of polluted cultivated		%	91	The expected nature of the indicators issued by the state	
18		land Risk control rate of soil pollution in key construction land		%	–	The expected nature of the indicators issued by the state	
19		Disposal of industrial hazardous		%	–	Safety, controllability and predictability	
20		wastes Harmless disposal rate of medical		%	–	100	Anticipation
21	pollute Prevention	Pollution put	waste Chemical oxygen demand emission	10,000 tons -	10,000 tons	The state issues binding indicators	
22			reduction Ammonia nitrogen	- 10,000 tons -		The state issues binding indicators	
23			emission reduction Nitrogen oxide			The state issues binding indicators	
24			emission reduction Volatile organic compound			The state issues binding indicators	
25		Living Pollution water	emission reduction Urban domestic sewage	%	–	70	Anticipation
26			centralized collection rate Township sewage	%	–	100	Anticipation
27			treatment facility coverage rate Rural domestic sewage treatment rate	%	22.1	35	Anticipation
28		Life garbage Rubbish	Resource utilization rate of urban domestic waste	%	–	60	Anticipation
29		Black and smelly water Body elimination	Built-up areas of prefecture-level cities (including Jishou City)	%	98.4	Consolidate and improve, maintain Long-term control	Anticipation
30		Proportion	County-level urban built-up area	%	–	and long-term cleaning basically eliminate the expectation	

Note: ÿ Wetland protection rate: from the "Fourteenth Five-Year Forestry Development Plan of Hunan Province".

ÿ The proportion of the ecological protection red line to the national land area: the final result shall be based on the optimization, adjustment and boundary survey and standardization results of the ecological protection red line organized by the Provincial Department of Natural Resources.

3. Striving for green, low-carbon and circular development

1. Optimize the national land space protection pattern.

Implement the main functional area strategy. Adhere to the priority of protection, based on the carrying capacity of resources and environment, reasonably divide the urbanization development area, agricultural product production area, and key ecological functional area, give play to the comparative advantages of various regions, implement the differentiated allocation of national space resources, and promote the formation of a main function with obvious advantages and advantages.

We will establish a new pattern of land space development and protection with complementary advantages and high-quality development, and protect permanent basic farmland and ecological space. We will promote ecological functional areas to focus on protecting the ecological environment and providing ecological products, promote major agricultural production areas to enhance agricultural production capacity, implement agricultural non-point source prevention and control and agricultural land

Risk management and control, promote intensive green and low-carbon development in urban development areas, promote urban ecological restoration, and build resilient, green, low-carbon, sponge cities.

Strengthen the zoning control of land space. Coordinate the demarcation of three control lines: ecological protection red line, permanent basic farmland and urban development boundary, reasonably arrange production, living and ecological space, form a scientific, moderate and orderly land space layout system, and reduce the occupation of natural space by human activities.

In principle, no human activities are allowed in the core areas of nature reserves within the ecological protection red line, and development and production construction activities are strictly prohibited in other areas. Under the premise of complying with current laws and regulations, only limited human activities that do not damage ecological functions are allowed, except for major national strategic projects, to ensure ecological

The function, area and nature of permanent basic farmland shall not be reduced. The protection of permanent basic farmland shall be strengthened, and the permanent basic farmland that is subject to detailed investigation and strict control of soil pollution shall be verified, rectified and supplemented to ensure that the area is not reduced, the quality is improved and the layout is stable. Urban development areas should fully consider the carrying capacity of resources and the environment.

We should rationally determine the development layout, structure and scale, and guide key industries to optimize their layout in areas with sufficient environmental capacity and good diffusion conditions.

Promote regional green development. Ensure the ecological safety of "one river, one lake, three mountains and four waters", and promote the coordinated development of the "Changsha-Zhuzhou-Xiangtan, Dongting Lake, Southern Hunan, and Greater Western Hunan" regions. The Changsha-Zhuzhou-Xiangtan region will accelerate the promotion of ecological environment co-construction and high-quality development, strengthen joint prevention and control of the atmosphere, and strengthen pollutant emission control and

Environmental risk prevention and control; Dongting Lake region focuses on upgrading agricultural structure, reducing agricultural non-point source pollution, optimizing chemical industry layout, improving water environment capacity, and strengthening wetland biodiversity protection; Southern Hunan region strictly controls the expansion of new heavy metal-related production capacity in the construction of industrial transfer demonstration areas.

Zhang, optimize the industrial layout, accelerate the transformation and upgrading of the industry, and speed up the solution of historical pollution problems; the Greater Western Hunan region strictly prohibits the transfer of high-energy-consuming and high-emission industries, takes the ecological industry as the orientation to undertake industrial transfer, maintains and strengthens the ecological safety barrier, strengthens ecological governance and restoration, and strengthens the "four

"Water" source pollution prevention and control and heavy metal pollution control in the river basin.

(ii) Promote the formation of green production methods.

Promote the green transformation of industrial structure. Accelerate the construction of green manufacturing system, continue to promote the construction of emerging industrial advantage chains and "3+3+2" key industrial fields, focus on carbon peak and carbon neutrality goals, and make breakthroughs in pollution control, comprehensive resource utilization, advanced energy storage, fuel cells, carbon capture and utilization, and sealing.

Breakthroughs in a number of key technologies will be made in the areas of inventory, etc. Outdated production capacity will be eliminated in accordance with laws and regulations using comprehensive standards, and new production capacity in industries such as coal, steel, cement, electrolytic aluminum, and flat glass will be strictly prohibited without approval. Resolutely curb the blind development of "two high" projects, comprehensively sort out and investigate "two high" projects under construction, and scientifically and effectively implement them.

We will promote the planned projects in an orderly manner, strictly implement the requirements for regional reduction of pollutant emissions, and resolutely stop approving and stopping construction of projects that do not meet the requirements. We will carry out comprehensive pollution reduction and carbon reduction management in industries such as coal power, petrochemicals, chemicals, steel, non-ferrous metallurgy, and building materials. We will formulate a provincial clean production audit implementation plan and deepen

The mandatory clean production audit will be promoted in the energy, metallurgy, coking, building materials, nonferrous metals, chemicals, printing and dyeing, papermaking, API, electroplating, agricultural and sideline food processing, industrial coating, packaging and printing industries. By 2025, all the requirements of the mandatory clean production audit program will be implemented, and key industries will be promoted.

Complete the upgrade and transformation of restricted capacity equipment. Actively promote the upgrading and transformation of industrial clusters such as building materials, chemicals, casting, printing and dyeing, electroplating, and processing and manufacturing, improve the level of industrial intensive and green development, and actively explore pilot projects for clean production audits in industrial parks and enterprise clusters.

Promote the continuous optimization of the energy structure. Optimize the energy structure, build a clean, low-carbon, safe and efficient modern energy system, control the total consumption of fossil energy, reasonably control the total consumption of coal, and improve the clean utilization rate of coal. During the "14th Five-Year Plan" period, coal consumption will basically peak, forming a non-fossil-based energy system.

We will further improve the province's oil and gas network, deepen the "Gasification of Hunan Project", and basically achieve "all-county access and full coverage" of natural gas. We will increase the introduction of high-quality energy from outside the province, such as "external electricity into Hunan" and "shale gas into Hunan", and accelerate the promotion of wind power, photovoltaic

The development of new energy, mainly power generation, will be coordinated to develop high-quality clean energy such as hydropower, hydrogen energy, geothermal energy, and biomass. By 2025, we will strive to reduce the proportion of coal consumption in the province to about 52%, strive to increase natural gas consumption to 10 billion cubic meters, and increase the proportion of non-fossil energy consumption to 23%.

We will upgrade and transform coal-fired power units, and gradually phase out coal-fired power units below 300,000 kilowatts in the Changsha-Zhuzhou-Xiangtan area. We will implement clean energy substitution for terminal energy, accelerate electrification in industries such as industry, construction, and transportation, promote clean energy substitution, gradually improve rural energy use structure, and advocate the use of solar energy, stone, and other energy sources.

Clean energy such as liquefied petroleum gas, electricity, and biogas.

Promote the continuous optimization of the transportation structure. Give full play to the water transport resource endowment of "one river, one lake and four waters" and the advantages of railway transportation that "connects the south and the north, connects the east and the west", promote the "road to rail, road to water" of medium and long-distance transportation of bulk goods and containers, and realize the optimization of "railway when appropriate, road when appropriate, and water when appropriate".

Combine, reduce road transport volume, increase rail and water transport volume. Strengthen the control of diesel truck bulk cargo port-to-port transport, gradually restrict and prohibit long-distance bulk cargo port-to-port transport by car, cultivate rail and water cargo transport market, and promote bulk cargo port-to-port transport to rail.

The proportion of green transportation for bulk cargo, and the proportion of railway and water freight volume will continue to increase. Gradually complete the elimination of old gasoline vehicles (national second-stage emission standards and below) and old diesel vehicles (national third-stage emission standards and below). By 2025, basically

Complete the elimination of old gasoline vehicles and 80% of old diesel vehicles.

Promote the green development of agricultural production. Guide the rational layout of livestock and poultry breeding and promote the combination of planting and breeding. Increase support for the resource utilization of livestock and poultry manure and strengthen the construction of livestock and poultry waste treatment facilities. Promote scientific fertilization, pesticide application, organic fertilizer instead of chemical fertilizer, and orderly promote the integration of water and fertilizer

Development. Promote the efficient utilization of agricultural and forestry wastes such as crop straw, livestock and poultry manure, forestry waste, and agricultural product processing by-products. Support townships in building a recycling system for waste agricultural materials such as waste agricultural films, fertilizer and pesticide packaging, irrigation equipment, and agricultural machinery. Promote the recycling of livestock, poultry, fish, grain,

Promote the coordinated development of vegetables, fruits and tea, promote the circular development of planting, breeding, agricultural product processing, biomass energy, and ecological tourism, and encourage the integrated development of the primary, secondary, and tertiary industries.

Promote efficient recycling of resources. Strengthen the management of the entire process of water and energy use in industrial production, improve the efficiency of water resources and energy utilization, strictly implement the total amount and intensity management of water and energy use, and develop an industrial energy efficiency and water efficiency "leader" system. Promote the circular transformation of industrial parks and encourage enterprises to

Circular production and industrial circular combination, build a public platform for resource sharing, waste treatment, and efficient services, promote the comprehensive utilization of industrial waste resources, energy cascade utilization, and water resource recycling. Accelerate the improvement of the market-based charging mechanism for the coordinated disposal of urban waste, and promote the establishment of an "interconnected

We will adopt the "Network + Recycling" model for recycling waste resources, make full use of and improve the reverse logistics recycling system of home appliance production and distribution enterprises, and establish a sound network for recycling waste home appliances. We will strengthen the standardized management and environmental supervision of enterprises dismantling and utilizing waste electrical and electronic products, scrapped motor vehicles, and waste lead-acid batteries, and improve the recycling of waste resources.

Build a modern "urban mineral" base. Improve the remanufacturing level of automotive parts, engineering machinery, machine tools, etc., and promote the high-quality development of the remanufacturing industry. Carry out benchmarking and improvement actions for key energy-consuming industries and product resource efficiency, and establish a unified green product standard, certification, labeling and other systems.

(III) Advocate a green and low-carbon lifestyle.

Practice a green lifestyle. Promote the establishment of a green product standard certification and labeling system, improve the effective supply of green and low-carbon products in key areas of daily life, encourage green consumption, and actively guide the purchase and use of green products. Promote green and paperless office; resolutely stop food waste;

We actively practiced the "Clean Plate Campaign", encouraged energy conservation and water conservation, reduced excessive packaging, and implemented waste reduction and classification; promoted greening of community infrastructure, rationally planned and constructed public green spaces; promoted green travel, encouraged the public to reduce the intensity of private car use; encouraged green consumption, and actively

Guide the purchase and use of green products; promote green building construction methods, and implement energy-saving renovation of existing residential buildings. Carry out creation activities such as energy-saving offices, green families, green schools, green communities, green travel, green shopping malls, and green buildings, and promote the "Citizen Ecological Environment Behavior Code"

(Trial Implementation)" to promote a green lifestyle and create a social atmosphere that advocates green living.

Continue to promote the whole chain of plastic pollution control. Establish and improve the coordination and promotion mechanism and responsibility implementation mechanism for plastic pollution control, and include plastic pollution control in the battle against pollution and the scope of provincial ecological and environmental protection supervision. Strengthen supervision and law enforcement, completely ban the import of waste plastics, and promote the use of degradable

Plastic products and plastic substitutes. Regularly clean up plastic waste in key areas such as rivers and lakes, coastlines, and beaches. Vigorously publicize and guide, encourage and support plastic-related institutions and enterprises to build cooperation platforms, formulate industry standards, carry out pilot demonstrations, strengthen scientific research, and cultivate new business forms and models.

Mode.

Strengthen the classification management of domestic waste. Implement the domestic waste classification system and build a domestic waste management system with classified delivery, collection, transportation and disposal. By 2025, the central cities of the province (prefectures) will basically have built a domestic waste classification and treatment system. Promote the construction of a domestic waste classification system.

The waste incineration power generation project has been launched to promote the transformation of the terminal treatment mode of domestic waste from landfill-based to diversified treatment modes with incineration as the main treatment mode; areas with daily domestic waste removal volume exceeding 300 tons in cities have basically achieved zero landfill of original waste, and the level of domestic waste reduction, resource utilization and harmlessness has been comprehensively improved.

By 2025, the harmless disposal rate of urban domestic waste will reach 100%, and the resource utilization rate of urban domestic waste will reach 60%.

Promote the resource utilization of construction waste. Promote the establishment of a standard system for the resource utilization of construction waste, form independent intellectual property rights in the treatment and disposal of construction waste, research and development of recycled materials, production and utilization, and break through the technical barriers between construction waste and recycled materials. Promote the establishment of provincial construction

By 2025, the construction waste resource utilization rate will reach more than 50%, 2-3 provincial construction waste resource utilization demonstration cities, and more than 10 construction waste resource utilization bases will be built.

Demonstration project, building 30 construction waste resource utilization bases.

Create a peaceful and harmonious living environment. Strengthen the management of acoustic environment functional areas, carry out assessment and adjustment of acoustic environment functional areas, and install automatic noise monitoring systems in acoustic environment functional areas in cities at prefecture level and above. Real estate development projects should fully consider the impact of ambient noise in the surrounding areas on the living environment of residential areas.

The noise reduction effect should be reduced by reasonably defining the noise prevention distance and clarifying the planning and design requirements. The sound insulation performance requirements of buildings should be raised, and a system for the acceptance and publicity of the sound insulation performance of newly built residential buildings should be established. Strict approval should be implemented for nighttime construction and the system should be made public. The use of low-noise construction equipment and processes should be encouraged, and the management of nighttime construction should be strengthened.

Incorporate industrial noise into the pollution permit management, and severely punish industrial noise emission exceeding the standard that disturbs the public. Strengthen daily supervision and centralized rectification of social life noise in cultural entertainment and commercial operations. Advocate the formulation of civilized conventions in public places and community noise control regulations, and encourage the creation of quiet communities.

Areas and other quiet rest spaces.

(IV) Actively respond to climate change.

Make every effort to promote carbon peak action. Promote high-quality economic development and high-level ecological environment protection by reaching carbon emission peak. Formulate Hunan Province's carbon dioxide emission peak action plan, clarify the peak target, roadmap and supporting measures, promote the preparation of peak plans for cities and prefectures, Changsha, Zhuzhou, Xiangtan, etc.

Cities will take the lead in achieving peak carbon dioxide emissions; promote key areas such as energy, industry, transportation, and construction to formulate peak action plans; promote key industries such as steel, building materials, nonferrous metals, chemicals, petrochemicals, and electricity to propose clear peak targets and formulate peak action plans; encourage large enterprises to

Formulate a peak action plan and implement pollution reduction and carbon reduction demonstration projects. Continue to promote low-carbon product certification and promote the application of low-carbon technologies.

Control greenhouse gas emissions. Upgrade the production processes of key industries such as steel, building materials, and chemicals to control greenhouse gas emissions from industrial processes. Promote raw material substitution technologies for cement production, and encourage coal-fired power, steel, petrochemicals and other industries to carry out full-process carbon dioxide emission reduction demonstration projects. Vigorously develop low-carbon

Transportation: By 2025, the reduction rate of carbon dioxide emissions per unit of transport turnover of operating vehicles and ships will meet national requirements, promote energy-saving and new energy vehicles, and accelerate the construction of charging infrastructure. Comprehensively promote green and low-carbon buildings, vigorously develop passive ultra-low energy consumption buildings, and promote green

We will encourage the use of prefabricated buildings in central urban areas where conditions permit, continue to promote green and energy-saving renovation of old residential buildings and public buildings, improve energy-saving design standards for new buildings, strengthen energy consumption monitoring and low-carbon operation management of public buildings, and by 2025, the area of newly completed green buildings in cities and towns will be

The proportion of newly completed residential buildings to the total area reached 75%, and the proportion of newly started prefabricated buildings in urban areas to the newly built building area reached 40%. Implement emission control of fluorinated greenhouse gases such as perfluorocarbons and nitrous oxide, promote sulfur hexafluoride replacement technology, control methane and oxidative gases in farmland and livestock and poultry breeding, and reduce the emission of fluorinated greenhouse gases such as perfluorocarbons and nitrous oxide.

Nitrous acid greenhouse gas emissions, and strengthen methane emission control and recycling from sewage treatment plants and landfills.

Actively respond to climate change. Strengthen macro-strategic coordination, promote the inclusion of climate change response work into economic and social development plans, and actively carry out climate change response actions in key areas such as agriculture, forestry, water resources, and infrastructure. Scientifically formulate special plans for responding to climate change and jointly promote

We will implement the goals and tasks of controlling greenhouse gas emissions and adapting to climate change, coordinate the planning of policy measures and key projects that are conducive to promoting green and low-carbon development in the economy, energy, and industry, and implement "dual control" of carbon dioxide emission intensity and total amount.

Lay a solid foundation for responding to climate change. Prepare a greenhouse gas emission inventory, improve the carbon emission accounting and verification system for regions, enterprises, and projects, and build a greenhouse gas collaborative governance system. Strengthen the monitoring and data reporting of greenhouse gas emissions by key enterprises, and promote the data collection and verification of greenhouse gas emissions by enterprises and institutions.

The unified collection, mutual supplementation and cross-checking of data will be implemented. Key units of greenhouse gas emissions will be included in the ecological environment supervision and law enforcement system. A consultation and sharing mechanism for basic data on climate change will be established to strengthen information sharing on high-energy-consuming and high-emission projects.

Promote the construction of low-carbon pilot demonstration projects. Strengthen the pilot projects of low-carbon cities, low-carbon industrial parks, and climate-adaptive cities, and improve the pilot assessment and evaluation system. Support places with good foundations to explore near-zero carbon emissions and carbon neutrality pilot projects. Select typical cities and regions to conduct air quality assessments.

We will carry out pilot demonstrations for the coordinated control of air pollutants and greenhouse gases in the steel, building materials, nonferrous metals and other industries. We will promote the organic integration of ecological civilization demonstration creation, low-carbon pilot projects, and climate change adaptation pilot projects. We will actively participate in the national climate investment and financing pilot projects.

We will promote major scientific and technological innovation and engineering demonstration, and make climate change response a key area of ecological and environmental science and technology development. We will encourage the development, demonstration and promotion of technologies related to the coordinated reduction of greenhouse gases and pollutants. We will actively participate in the construction of the national carbon market and strengthen key emission reduction projects.

Supervision and management work such as unit data submission, verification, quota settlement and fulfillment.

(V) Strictly control ecological and environmental access.

Strictly implement the overall ecological and environmental management requirements of Hunan Province's "three lines and one list", and implement the ecological protection red line, environmental quality bottom line, resource utilization upper line and ecological environment access list as hard constraints to environmental management units.

Differentiated ecological and environmental access management shall be implemented for environmental control units based on resource endowment, economic and social development. The connection between the "Three Lines and One List" and national spatial planning shall be strengthened. Regional resource development, industrial layout and structural adjustment, urban construction, and site selection of major projects shall be based on the "Three Lines and One List".

Environmental control units and ecological environment access lists are used as important bases to strengthen the ecological environment access management of industrial parks at or above the provincial level. Promote the sharing of data systems such as "three lines and one list" and pollutant discharge permits, environmental assessment approvals, environmental monitoring, and environmental law enforcement, and refine the data support system and zoning of "three lines and one list"

Management and control requirements.

Strengthen planning environmental impact assessment. Strictly implement the ecological environment source prevention system with the environmental impact assessment system as the main body, guide economic and social development with national spatial planning, regional planning, and industry development planning, and comprehensively promote planning environmental impact assessments for key areas, key river basins, and key industries. Planning preparation

We must fully consider bottom-line constraints, space control, total volume control and ecological environment access, and coordinate regional spatial layout and ecological safety pattern. Strictly review the relevant comprehensive plans and special plans for industry, energy, etc. related to the "two highs" industries, strictly control the development scale of the "two highs" industries, optimize the planning layout, and strictly control the development scale of the "two highs" industries.

Strengthen the guidance and constraints of planning environmental assessment on the environmental assessment of construction projects and promote the implementation of planning environmental assessment results.

Fully implement the pollutant emission permit system. Promote the construction of a fixed pollution source supervision system with the pollutant emission permit system as the core, achieve full coverage of pollutant emission permits for fixed pollution sources, promote full coverage of industrial solid waste and soil environmental elements, and explore the inclusion of carbon emissions in the management of pollutant emission permits.

Pollutant emission permits are used to allocate, supervise and assess the total amount of pollutant emissions from enterprises and institutions. A daily ecological and environmental supervision and law enforcement system based on pollutant emission permits will be established to implement the "one-permit" management of pollutant emission permits. The effective integration of the pollutant emission permit system and the environmental impact assessment system will be promoted.

Promote the full closed-loop management of environmental impact assessment, pollutant discharge permits, supervision and law enforcement for key industries and enterprises. Continue to update the renewal or registration of pollutant discharge permits.

IV. Fighting the tough battle against pollution

1. Fight hard to protect clear water.

Strengthen the protection and supervision of drinking water sources. Optimize the layout of drinking water sources and promote the integration of urban and rural water supply. Strengthen the construction of backup and emergency water sources and improve emergency water supply capabilities. Continue to promote the demarcation of centralized drinking water source protection areas. Strengthen the environmental management of drinking water source protection areas,

Consolidate the achievements of standardized construction of water source protection areas at the county level and above, complete an annual survey and assessment of the environmental status of drinking water source protection areas, apply remote sensing monitoring and manual inspection methods, timely identify emerging environmental problems in drinking water source protection areas, solve environmental problems in a timely manner, and maintain drinking water.

Water source and water quality safety. In 2021, the province will basically complete the demarcation of centralized water source protection areas with more than 1,000 people, the standardized construction of centralized water source areas with more than 1,000 people at the township level, and the rectification of outstanding environmental problems; in 2023, the province will basically complete the standardized construction of centralized water source protection areas with more than 1,000 people and the rectification of outstanding environmental problems.

Strengthen the monitoring of water quality of centralized drinking water sources, and include the "thousand tons per 10,000 people" centralized drinking water sources in routine monitoring, establish a drinking water information management platform, and improve the environmental emergency management mechanism for drinking water sources. By 2022, the centralized drinking water sources at the county level and above, and in townships and rural areas will be completed.

Conduct environmental risk assessments at water sources and record emergency plans for sudden environmental incidents, conduct regular emergency drills at water sources, and strengthen emergency preparedness, early warning, and emergency response to sudden environmental incidents. Establish a water source risk assessment and water quality early warning and forecasting system.

Strengthen water pollution control in key river basins. Integrate water functional zones and water environment functional zones, and clarify water quality protection targets at all levels of control sections. Implement the Dongting Lake total phosphorus control and reduction action, strengthen industrial, agricultural, and domestic pollution control, and continuously reduce total phosphorus pollution in the lake area and the lake basin.

The total amount of pollutant emissions will be reduced, the connection between rivers and lakes will be strengthened, the ecological water volume in lake areas will be guaranteed, and the water environment capacity will be increased. For the water areas in the Xiang-Zi-Yuan-Li main stream and key lakes and reservoirs where the water quality is not stable and does not meet the standards, a plan for meeting the standards will be formulated and implemented, and the water areas that do not meet the standards will be pushed to meet the standards within a time limit. People's governments at or above the county level will be encouraged to take measures to increase the water quality of their jurisdictions to

The water quality of the exit section of the district has been raised to the surface water Class II standard. We will continue to strengthen the protection of good water bodies and include lakes and reservoirs with important protection significance into the protection scope of good water bodies. We will carry out in-depth treatment of basin-wide heavy metal pollution caused by mine water gushing and historical waste residues, focusing on solving the water pollution of antimony, thallium, etc.

By 2024, the surface water system water quality control unit will stably meet the assessment target. We will continue to fight the battle to treat urban black and odorous water bodies, achieve long-term cleanliness of black and odorous water bodies in urban built-up areas at the prefecture level and above, and basically eliminate black and odorous water bodies in county-level urban built-up areas.

Deepen water pollution control in key areas. Make up for the shortcomings of urban and rural sewage collection and treatment facilities, strengthen pollution control of domestic sources, improve urban sewage pipe network construction, achieve full coverage of sewage pipe network in built-up areas, renovate old and damaged pipe networks and inspection wells, and systematically solve the problem of pipe network leakage. By 2025

By 2020, the gaps in domestic sewage collection and treatment facilities in urban villages, old urban areas and urban-rural fringe areas will be basically eliminated, the centralized collection rate of urban domestic sewage will reach 70%, and sewage treatment facilities will be fully covered in township government locations across the province. Focusing on enterprises and industrial clusters, promote sewage treatment in industrial parks

Classify and manage facilities, upgrade and renovate them in phases, implement special rectification actions for industrial parks at or above the provincial level, achieve full coverage of sewage pipe networks in industrial parks at or above the provincial level, collect all sewage, ensure stable and up-to-standard operation of centralized sewage treatment facilities, ensure normal online monitoring of inlet and outlet water quality, and standardize the setting of industrial parks.

The district will centralize the discharge outlets of sewage treatment facilities and establish a "one park, one file" for water environment management in the park. We will strengthen wastewater treatment in enterprises involved in heavy metal industries and promote the total amount of ammonia nitrogen and total phosphorus emissions in key industries.

Strengthen the management of the Yangtze River main and tributary systems. According to the requirements of the "Implementation Plan for the Relocation and Reconstruction of Chemical Enterprises Along the Yangtze River in Hunan Province", it is strictly prohibited to build or expand chemical industrial parks and chemical production projects within 1 km of the river bank; it is strictly prohibited to expand existing compliant chemical industrial parks along the river within 1 km of the river bank;

Chemical production enterprises that meet safety and environmental protection standards can be gradually relocated to the back of the river due to production needs, and the relocation and transformation of chemical enterprises along the river will be completed by the end of 2025. Comprehensively strengthen the investigation, rectification and supervision of sewage outlets into the river, and complete the Hunan section of the Yangtze River main stream, the Xiangjiyuanli main stream and important tributaries into the river by 2023.

Investigate the sewage outlets, establish a list of sewage outlets into rivers, and initially build a unified basin sewage outlet information management system. Complete the investigation of sewage outlets in the entire basin by 2025 and build a basin sewage outlet information management system. Complete the construction of a regional zoning system for sewage outlets into rivers, and clearly define areas where sewage outlets are prohibited or restricted.

Strengthen pollution prevention and control of ships and ports, optimize the layout of ports, clean up illegal ports, upgrade pollution prevention and control facilities for legal ports that do not meet environmental standards, promote the construction of green ports and green ports; improve ship life

The construction of sewage, garbage, and oily water receiving and transfer facilities should be promoted to effectively connect the receiving facilities with urban public transfer facilities. The ports and terminals in the Hunan section of the Yangtze River mainstream should build fixed receiving facilities for domestic sewage from ships at the port. The application of the joint supervision and service information system for ship water pollutants should be promoted to form a

Long-term mechanism for the prevention and control of pollution from ships and ports. Rationally arrange sand and gravel receiving terminals to guide the orderly development and application of river sand and gravel resources. Continue to carry out special investigation and rectification actions on "three phosphorus" in the Yangtze River Basin.

Strengthen the protection and utilization of water resources. Strengthen the connection between rivers and lakes, ensure the ecological water volume of rivers and lakes, restore the ecological functions of rivers and lakes, and focus on implementing the water network connection of six major areas, including the four-mouth water system of Dongting Lake, the East Dongting Lake area, the tail of Xiangzi, the Yuannan area, the Yuanli area, and the Songli area. Scientifically determine the ecological

The ecological flow targets of the Xiang-Zi-Yuan-Li main stream and important sections of important tributaries will be determined; water volume dispatching will be implemented for projects such as reservoirs and hydropower stations in accordance with the requirements for ensuring ecological flow during the dry season. The small hydropower remediation will be completed in an all-round way, and the small hydropower stations that are to be withdrawn will be withdrawn on schedule; the ecological flow of the small hydropower stations that are to be retained will be implemented.

Build a network of small hydropower and river and lake ecological water monitoring stations, and connect monitoring data to the provincial river and lake ecological water monitoring system; establish an ecological water monitoring, early warning and information release mechanism for important rivers and lakes.

Promote the recycling of urban life, industry, agriculture and rural sewage resources. On the basis of stable and standard discharge of sewage treatment plants, water-deficient areas with small environmental capacity and large fluctuations in water environment quality should give priority to converting standard discharge tail water into usable water resources.

The utilization level of poultry manure and fishery tailwater resources has increased significantly.

Strengthen water ecology protection and restoration. In accordance with the principle of "where there are rivers, there are fish, where there are fish, there are grasses", promote ecological expansion and carry out water ecology restoration. Strengthen the management of river and lake buffer zones, protect natural wetland resources, meet the ecological water requirements of important wetlands, repair damaged riverside, lakeside, and river mouth wetlands, and continue

Clear out production activities that damage the aquatic ecology. Implement vegetation restoration on the sites where the European and American poplars were cleared in Dongting Lake, and strengthen the construction of artificial wetlands at important river (lake) entrances. Continue to promote the ecological construction of important water conservation areas, strengthen the restoration and construction of ecological buffer zones at the tail of rivers (lakes), and carry out the "three fish fields" and

Protect and restore migratory channels, and build wildlife habitats and food sources. Strictly implement the "10-year fishing ban" requirement in the Yangtze River, carry out rescue protection actions for endangered aquatic species represented by Chinese sturgeon, finless porpoise, and carmine fish, carry out artificial breeding and population restoration, and promote aquatic biodiversity.

Sexual protection and restoration.

Column 1: Key "Clear Water" Projects

Dongting Lake total phosphorus control and reduction action. Implement the Dongting Lake total phosphorus control and reduction action, strengthen industrial, agricultural, and domestic pollution control, and continuously reduce the total amount of total phosphorus pollutant emissions in the area around the lake and the basin entering the lake. Strengthen the connection between rivers and lakes, ensure the ecological water volume in the lake area, and improve the water environment capacity. Further increase the efforts in wetland protection, lakeside and riverside ecological buffer zone construction, and effectively improve the self-purification capacity of the environment.

Pollution control, bank control and fishery control in the Yangtze River mainstream and tributaries. (1) Rectification of prominent ecological and environmental problems in the Yangtze River Economic Belt. (2) The "4+1" project for urban sewage and garbage treatment, chemical pollution control, agricultural non-point source pollution control, ship pollution control and tailings pond pollution control in the Yangtze River Economic Belt. (3) Control of black and odorous water bodies along the river, comprehensive control of coal mining subsidence areas, and systematic control projects for key river basins. Investigation and rectification of sewage outlets into the Yangtze River mainstream and major tributaries, and implementation of a project to standardize sewage outlets into the river. (4) Bank control of the Yangtze River mainstream and tributaries, implementation of illegal dock control and shoreline ecological restoration; chemical park and chemical enterprise control projects. (5) Implementation of the "ten-year fishing ban" requirements for the Yangtze River, biodiversity protection and restoration projects. (6) Control and water ecological restoration of key small watersheds, and mine ecological restoration projects.

Pollution control projects in key areas. (1) In-depth treatment of industrial water pollution and comprehensive environmental improvement of provincial and above industrial parks. (2) New construction and renovation of sewage pipe networks (3) Treatment of black and odorous water bodies in counties and cities.

Centralized drinking water source and good water body protection project. (1) Standardized construction and improvement project of centralized drinking water source with a capacity of 1,000 tons and serving 1,000 people or more. (2) Protection project of lakes and reservoirs with good water quality such as Dongjiang Lake and Shuifu Temple. (3) Beautiful river and lake creation project.

(ii) We will continue to fight hard to protect blue skies.

Promote the coordinated control of PM2.5 and ozone. Promote the continuous decline of urban PM2.5 concentration and effectively curb the increasing trend of ozone concentration. Formulate an action plan to strengthen the coordinated control of PM2.5 and ozone and continuously improve air quality, and clarify the control goals, roadmap and timetable. Strengthen the control of Changsha, Zhuzhou and Xiangtan and their transmission

Air pollution control in cities along the transmission channel will be strengthened, and differentiated and refined coordinated control of PM2.5 during the special protection period and ozone in summer will be strengthened. Research on coordinated control of PM2.5 and ozone will be strengthened, and pilot work on coordinated control of PM2.5 and ozone will be carried out in typical areas. Coordinated efforts will be made to promote the emission reduction of greenhouse gases and major air pollutants.

We will strengthen energy conservation and environmental protection constraints, and while drastically reducing major air pollutants, we will also work together to further reduce greenhouse gas emissions. We will strengthen the prevention and control of ammonia emissions in the breeding industry and encourage large-scale farms to implement ammonia emission controls.

Strengthen the in-depth treatment of NOx in key industries . Promote the upgrading and transformation of treatment facilities in the sintered brick and tile industry, eliminate the "double alkali method" integrated desulfurization and dust removal technology, and by 2025, sintered brick and tile enterprises will complete the transformation of high-efficiency desulfurization and dust removal. Promote cement clinker production enterprises to adopt technologies such as staged combustion, and

Prepare high-efficiency dust removal and denitrification facilities, implement in-depth treatment of nitrogen oxides, and control NOx emission concentration below 100 mg/m3 by 2023. Orderly promote ultra-low emission transformation of the steel industry. By the end of 2023, ultra-low emission transformation of steel enterprises in the province will make significant progress. By 2025,

By the end of the year, all steel enterprises will have completed ultra-low emission transformation. Promote in-depth pollution control in industries such as glass, ceramics, casting, and nonferrous metals. Strengthen the operation control of pollution control facilities for self-provided coal-fired units to ensure ultra-low emission operation. Coking, cement, bricks and tiles, lime, refractory materials, nonferrous metal smelting

Strictly control unorganized emissions in industries such as refineries. Key gas emission enterprises should gradually cancel flue gas bypasses. If they cannot be cancelled for safety reasons, install online monitoring systems. Carry out low-nitrogen transformation of gas boilers.

Strengthen scientific management of VOCs in key industries. Focus on industries such as industrial coating, petrochemicals, chemicals, packaging and printing, and oil storage, transportation and marketing, and implement VOCs raw material substitution and emission control in enterprises. In accordance with the principle of "different industries and different policies for different industries", increase the use of raw and auxiliary materials with low VOCs content.

Promote the use of advanced production technology equipment to reduce unorganized emissions. Implement dual control of emission concentration and removal efficiency of key emission sources. Strengthen the comprehensive management of VOCs in the auto repair industry, increase the management of restaurant fume pollution, and promote county-level

The above cities have full coverage of restaurant fume control.

Strengthen joint prevention and control of vehicles, ships, oil, roads and ports. Vigorously promote the control of air pollution from ships, compulsorily scrap ships that have exceeded their service life in accordance with the law, and encourage the elimination of inland shipping ships that have been in use for more than 20 years. Strictly implement gasoline and diesel quality standards, strengthen oil product supervision and law enforcement, and by the end of 2025, the annual sales of

Gas stations that sell more than 5,000 tons of gasoline should install automatic monitoring equipment for oil and gas recovery and connect with the competent industry departments to crack down on illegal gas stations and non-standard oil production companies. Vigorously carry out supervision and inspection of newly produced motor vehicles, engines, and non-road mobile machinery, and the main vehicle (machine) model families should be inspected annually.

The sampling rate of the province reached over 80%, and unregistered or black smoke-emitting engineering machinery was basically eliminated. The shore power construction of major ports in the main tributaries of the Yangtze River was promoted, and the shore power utilization rate was gradually increased.

Strengthen the precise and scientific control of dust pollution. Strictly implement the "six 100%" dust prevention and control measures at housing construction and municipal infrastructure construction sites in built-up areas of cities above the county level, fully promote green construction, and include green construction in enterprise qualification and credit evaluation. Promote low-dust mechanization

Wet cleaning operations should be carried out to increase the intensity of flushing and cleaning of important sections such as city entrances and exits and urban-rural junctions. Slag trucks should be transported in hard covers and fully enclosed, and dust control should be strengthened in road greening areas. Dust control at dock operations should be strengthened. Coal, ore and dry bulk cargo terminals should be fully equipped with wind and dust suppression equipment.

For the construction of facilities, the terminal yard should adopt a closed method for storage.

Actively respond to heavy pollution weather. Strengthen regional cooperation with neighboring provinces, explore the establishment of an inter-provincial early warning and joint defense mechanism, build a three-dimensional prevention and control network, and promote the formation of a regional "unified planning, unified standards, and unified supervision" linkage system. Strengthen early warning in the Changsha-Zhuzhou-Xiangtan region and cities along the air pollution transmission channel.

Strengthen emergency response to heavy pollution weather, revise and improve and continuously update emergency response plans for heavy pollution weather, refine emergency emission reduction measures, and implement emergency emission reduction list management. Urge industrial enterprises to follow the "one factory, one plan" principle.

According to the requirements, formulate specific emergency response operation plans. Initiate emergency response measures in a timely manner according to the heavy pollution weather conditions, strengthen the inspection and evaluation of the implementation of emergency plans, and improve the effectiveness of emergency measures. Explore the control of moderate and light pollution weather.

Carry out fine particulate matter compliance actions. Seven cities that have met the standards, including Hengyang, Zhangjiajie, Yongzhou, Chenzhou, Loudi, Jishou, and Huaihua, have continued to reduce the level of fine particulate matter in ambient air and consolidate and improve the quality of the atmospheric environment. Seven cities that have not met the standards, including Changsha, Zhuzhou, Xiangtan, Yueyang, Yiyang, Changde, and Shaoyang, have continued to reduce the level of fine particulate matter in ambient air and consolidate and improve the quality of the atmospheric environment.

Cities that meet the standards should formulate and implement a plan for meeting the air quality standards within a specified period and announce it to the public, clarify the route to meeting the air quality standards and the key tasks of pollution prevention and control, and strengthen the management of the meeting process in accordance with the principle of tightening first and loosening later and continuous improvement. By the end of the 14th Five-Year Plan, we will strive to have more than three cities that meet the standards in the province.

city.

Strengthen the control of other air pollutants. Strengthen the environmental management of ozone-depleting substances and hydrofluorocarbons, and strengthen the prevention and control of odorous, toxic and harmful air pollutants.

Column 2 "Blue Sky" Key Projects

Joint air pollution prevention and control project in Changsha, Zhuzhou, and Xiangtan and cities along their air transmission corridors. Eliminate coal-fired boilers in Changsha, Zhuzhou, and Xiangtan and cities along their air transmission corridors, implement green transformation and upgrading and comprehensive remediation of key industries, clean energy substitution, centralized heating, clean use of coal, etc., carry out urban construction dust control, road cleaning, air quality early warning and forecasting

projects. **Provincial PM** 2.5 and ozone pollution coordinated control project. Carry out comprehensive control of industrial VOCs, in-depth control of industrial NOx, diesel engine emission control project. Non-point source pollution control project (dust pollution control, straw comprehensive utilization, restaurant fume comprehensive control, odor control).

Ultra-low emission transformation of non-electric industries. Orderly promote ultra-low emission transformation of the steel industry. By the end of 2023, Hunan Valin Xiangtan Iron and Steel Co., Ltd. will complete ultra-low emission transformation, and three steel companies including Hengyang Valin Steel Pipe Co., Ltd., Hunan Valin Lianyuan Iron and Steel Co., Ltd. and Lengshuijiang Iron and Steel Co., Ltd. will complete more than 80% of the ultra-low emission transformation tasks; by the end of 2025, all steel companies will complete ultra-low emission transformation.

(III) We will fight hard to protect the clean land.

Implement fine management of the soil environment. Continue to promote the investigation and rectification of key industries involving cadmium and other heavy metals around cultivated land, identify and investigate the causes of cultivated land pollution; focus on the vacated land used by mining enterprises and the land used for residential and public management and public service purposes, and carry out soil

Pollution status survey and risk assessment. Establish a polluted land database and information platform shared by the ecological environment, natural resources, housing and urban-rural development departments, and realize the "one map" information sharing and joint supervision of suspected polluted land, polluted land spatial information, and national spatial planning.

Strengthen the prevention of soil pollution sources. Promote the coordinated control between pollutants and the soil and groundwater environments, and continue to conduct investigations and assessments of the soil and groundwater environments around solid waste and hazardous waste storage sites. Strictly control the discharge of pollutants from heavy metal-related industries, and rectify the heavy metal-related

It is a solid waste left over from the mining area, and it is used to prevent and control soil pollution caused by the development of mineral resources. A list of key supervision units for soil pollution will be established and updated dynamically in a timely manner, and key supervision units will be urged to fully implement the soil environmental management system in accordance with the law. By the end of 2025, at least one soil and groundwater pollution hazard will be completed.

Investigate and formulate and implement rectification plans. Key soil pollution supervision units should purchase environmental pollution liability insurance in accordance with relevant national regulations. Key soil pollution supervision units are encouraged to implement pipeline and closed transformation according to local conditions.

Consolidate and improve the safety level of cultivated land. Strengthen the protection of priority protected cultivated land to ensure that the area does not decrease and the environmental quality does not decline; in areas where permanent basic farmland is concentrated, no new construction projects that may cause soil pollution shall be planned. Strengthen the control of polluted cultivated land and summarize the

The pilot projects on restoration of heavy metal-contaminated farmland and adjustment of crop planting structure should be used to establish and improve a long-term mechanism for the safe use of contaminated farmland, improve the technical guidelines for the safe use of contaminated farmland, and the recommended catalogue for adjustment of planting structure in strictly controlled areas, accelerate the demarcation of strictly controlled areas for rice, and strictly control the import and export of rice.

Grain and rice production; encourage the adoption of measures such as planting structure adjustment and returning farmland to forest and grassland to ensure the ecological restoration and sustainable and safe use of strictly controlled farmland; promote comprehensive supporting technologies such as variety substitution, fertilizer regulation, and soil conditioning according to local conditions, and continuously improve the safe use of polluted farmland.

In accordance with the Technical Guidelines for the Delineation of Strict Control Areas for Specific Agricultural Products (Trial Implementation), the soil environment quality category of cultivated land shall be adjusted in a timely manner based on routine soil environment quality monitoring, agricultural product sampling, and evaluation of the effects of governance and restoration.

Strictly manage the access to polluted land. Reasonably plan the use of polluted land, and fully consider the soil pollution risks of construction land when formulating relevant plans such as national land space. For plots that are required to conduct soil pollution status surveys according to law, complete surveys and risk assessments, and reasonably determine the use of land.

The land use management mechanism shall be strictly controlled. The land listed in the list of construction land soil pollution risk control and remediation shall not be used as residential, public management and public service land. The land that does not meet the requirements of the soil pollution risk assessment report shall be prohibited from construction and risk control and remediation.

Unrelated projects; plots that have not completed soil pollution status surveys or risk assessments are prohibited from entering the land use procedure. Reasonable planning of the time sequence for the redevelopment of polluted plots, and the optimization of the time sequence for the phased development of large-scale polluted plots. In principle, sensitive areas such as residential, public management and public services should be

Sensitive land should be developed later. For polluted land with prominent pollution, environmental sensitivity and high control difficulty, avoid developing and using it as high-function land.

Steadily promote soil control and restoration. Implement actions to protect and improve the quality of cultivated land, and carry out acid reduction and improvement of highly acidic soil. Focus on cadmium-contaminated cultivated land and promote pilot projects for the restoration of cultivated land soil pollution.

Strictly implement risk control and governance and restoration measures for the relocated and reclaimed land of hazardous chemical production enterprises. Strengthen research on soil pollution risk control and restoration technology, explore the "remediation factory" model for contaminated soil in Zhuzhou Qingshuitang, Xiangtan Zibugang and other areas, and build contaminated soil restoration

Disposal center. Implement key risk control for contaminated plots that are not yet developed and where treatment technology is not yet mature to prevent the spread of pollution. Strengthen risk control and remediation project supervision and promote the concept of green remediation.

Strengthen groundwater environmental protection. Conduct groundwater environmental surveys in groundwater drinking water source protection areas and recharge areas, implement standardized construction in designated groundwater drinking water source protection areas, and establish and improve priority control of pollution source cleaning in groundwater drinking water source recharge areas.

Conduct investigation and rectification of hidden dangers around groundwater environmental quality assessment points. Conduct groundwater environmental status survey and assessment of "one enterprise, one reservoir" and "two fields, two areas" (i.e. chemical production enterprises, tailings reservoirs, hazardous waste disposal sites, landfills, industrial clusters, and mining areas).

Groundwater environmental risk. Before the end of 2023, a number of groundwater environmental surveys and assessments will be completed for industrial clusters and hazardous waste landfills; before the end of 2025, a number of groundwater environmental surveys and assessments for other pollution sources will be completed. Based on the survey and assessment results, a groundwater environmental survey and assessment will be conducted.

and restoration pilot projects to control groundwater pollution.

Column 3: “Pure Land” Key Project

Investigation, assessment and remediation projects. (1) Comprehensive investigation of soil pollution status of land used by key industry enterprises. (2) Risk control or remediation of polluted plots in key areas. (3) Investigation of causes of heavy metal pollution in cultivated soil. (4) Treatment of mine water inflows from unowned mines containing heavy metals. (5) Construction of groundwater supervision capacity and demonstration of groundwater environmental status investigation, assessment, control and remediation.

Pilot project for the treatment of heavy metal polluted farmland. Take the heavy metal polluted farmland in Changsha, Zhuzhou and Xiangtan as the pilot area. Accurately classify the soil environmental quality of farmland, and implement classified and zoned management for priority protection, safe utilization and strict control farmland. Carry out safe utilization and strict control of farmland polluted by heavy metals, carry out demonstration and verification of blocking and control technologies for the current major pollution sources, compile source control technology guidelines, screen heavy metal polluted farmland treatment technologies, and establish demonstration bases.

(4) Deepen agricultural and rural environmental governance.

Promote rural domestic sewage treatment. Implement the rural domestic sewage treatment plan, focus on villages around environmentally sensitive areas, township government seats and central villages to promote rural domestic sewage treatment in stages, promote the extension of urban sewage treatment facilities and services to rural areas near towns, and coordinate the promotion of rural domestic sewage treatment.

By 2025, rural sewage treatment facilities will be basically fully covered in important ecological function areas such as drinking water source protection areas and nature reserves, the vicinity of Dongting Lake, along the Xiang-Zi-Yuan-Li main stream, and in administrative villages of control units of national and provincial sections that have not met the standards, and nearly all urban sewage treatment facilities will be covered.

The domestic sewage treatment rate in suburban rural areas has reached about 55%, and the domestic sewage treatment rate in rural areas across the province is not less than 35%.

Promote the treatment of rural domestic waste. Achieve basic coverage and stable operation of rural domestic waste collection and transfer facilities. Carry out on-site classification of rural domestic waste, give priority to promoting waste classification in suburban villages, and ensure that hazardous waste and kitchen waste are placed separately. Promote resource utilization suitable for rural characteristics.

By 2025, the proportion of administrative villages that collect and treat rural domestic waste by classification will reach 30%.

Strengthen the prevention and control of agricultural non-point source pollution. Demarcate the priority control areas for agricultural non-point source pollution, carry out pilot projects for comprehensive control and supervision of agricultural non-point source pollution, and build a "one-stop" monitoring network for agricultural non-point source pollution. Deeply promote new agricultural technologies, such as soil surveying and formula fertilization, organic fertilizers replacing chemical fertilizers, and water-fertilizer integration.

With the integration of agricultural production, integrated pest and disease control and green control technology as the core, we will promote the reduction and efficiency of chemical fertilizers and pesticides. We will carry out green planting and breeding cycle agriculture pilot projects and green control demonstration projects in the main grain-producing areas along the "one lake and four rivers" to improve the utilization rate of chemical fertilizers and pesticides. By 2025, the application of chemical fertilizers and pesticides for major crops will be reduced by 1.5%.

The amount used has continued to decrease, and the reduction target has reached the national requirements. We will promote the green compensation system for comprehensive utilization of straw and the construction of pilot counties for comprehensive utilization of straw, continuously improve the network for recycling and utilization of agricultural film and straw, promote the construction of the recycling and utilization industry chain of agricultural film and straw, and strictly prohibit open-air burning of straw. Implementation

The policy of returning straw to the fields and removing it from the fields will be supported, and detailed rules for the treatment of discarded agricultural films will be studied and formulated. By 2025, the agricultural film recycling rate and the comprehensive utilization rate of straw in the province will meet national requirements.

Strengthen the prevention and control of pollution in the breeding industry. Adhere to the principle of determining the number of livestock and poultry according to the land and the number of breeds, optimize the regional layout of livestock and poultry breeding on a county basis, scientifically plan the spatial layout of the breeding industry, determine the scale of livestock and poultry breeding according to the carrying capacity of the land, and gradually reduce the number of livestock and poultry breeding in areas and farms that exceed the carrying capacity of the land.

Focusing on large livestock and poultry breeding counties and scale farms, with agricultural organic fertilizer and rural energy as the main utilization direction, we will increase the county-wide promotion of livestock and poultry manure resource utilization, strengthen the supervision of livestock and poultry breeding below scale, encourage breeders to collect and utilize livestock and poultry manure in full, and actively promote

Economical and efficient technology model for the resource utilization of manure and sewage. Improve the level of integration of planting and breeding, focus on advantageous fruit, vegetable and tea production areas, core production areas, and well-known brand production bases, support and guide farmers and new business owners to build and apply organic fertilizers, guide national modern agricultural demonstration areas, national modern agricultural production bases, and national modern agricultural production bases.

We will take the lead in realizing the circular development of planting and breeding in the national agricultural green development pilot zone and the national agricultural green development pilot zone. We will optimize the spatial layout of aquaculture, adhere to ecological and healthy breeding, and carry out comprehensive treatment of aquaculture tailwater pollution with the Dongting Lake area as the focus. By 2025, the comprehensive utilization rate of livestock and poultry manure in the province will reach more than 80%.

In terms of the above, the problem of manure treatment and resource utilization in large-scale livestock and poultry farms has been basically solved.

We will carry out in-depth improvement of the human settlement environment. We will carry out rural human settlement environment improvement and upgrading activities in a county-wide manner, vigorously promote environmental improvement in poverty-stricken areas, support the masses in carrying out village cleaning and greening activities, and promote the construction of beautiful rural model villages in a hierarchical and categorized manner.

Improve the county-wide promotion mechanism, and optimize the reward and punishment indicators for rural human settlement environment improvement. Coordinate the implementation of rural black and odorous water treatment and rural water system comprehensive improvement, and give priority to the cause elimination of rural black and odorous water in important ecological function areas and water bodies with large areas, severe pollution and strong public complaints.

Investigate and carry out pilot projects to form a number of replicable and popularizable rural black and odorous water governance models, and explore the establishment of a long-term mechanism for rural black and odorous water governance. By 2025, 3,200 rural environmental governance villages will be completed, and the rural black and odorous water governance rate will reach 40%.

Improve the rural environmental governance mechanism. Implement the rural environmental governance management responsibilities of the four levels of city (state), county (district), township (town), and village, strengthen the linkage between departments and localities, incorporate rural domestic sewage and garbage disposal into village rules and regulations, and gradually form a mechanism for party and government agencies, enterprises, institutions, and the general public to govern the rural environment.

Create a good atmosphere for conscious participation in rural environmental protection. Improve the operation and management mechanism of rural domestic sewage treatment facilities and rural domestic garbage collection and transportation systems, and explore the establishment of a sewage treatment farmer payment system and a rural garbage treatment fee system. Encourage local governments to introduce policies to support the production, storage and transportation of organic fertilizers.

Improve market incentive mechanisms, study and formulate preferential policies for land and electricity use for organic fertilizer plants, large-scale biogas projects, and third-party treatment of livestock and poultry manure, promote the grid connection and operation of power generation from straw and livestock and poultry manure, and fully guarantee the purchase of electricity. Combined with the actual situation, increase subsidies for returning straw to the fields, and explore

Explore the establishment of a compensation mechanism for the consumption of ecological and organic agricultural products. Innovate the investment and financing mechanism for rural environmental protection, and provide key support for investment in rural environmental infrastructure projects and environmental governance funds through the issuance of guiding policy loans;

Measures should be taken to attract social capital to enter the construction of rural environmental protection infrastructure; and government procurement of services should be actively explored.

Column 4: Agricultural and rural ecological environment improvement projects
Rural domestic sewage treatment step-by-step promotion project. According to the rural domestic sewage treatment plan and national assessment requirements, the province has completed 3,200 administrative villages. Live sewage treatment. Comprehensive prevention and control of agricultural and rural non-point source pollution. Focusing on the Dongting Lake area, taking into account suburban areas, hilly and mountainous areas, we will carry out the construction of rural non-point source pollution comprehensive prevention and control demonstration areas, focusing on the construction of regional agricultural and rural scale aquaculture tailwater treatment, livestock and poultry manure comprehensive treatment and resource utilization and other non-point source pollution prevention and control projects, forming a number of replicable and popularizable pollution prevention and control models. Demonstration project for the treatment of rural black and odorous water bodies. Use satellite remote sensing monitoring technology to conduct a comprehensive investigation of rural black and odorous water bodies across the province, and form a black and odorous water body treatment project. List; carry out pilot work on the treatment of black and odorous water bodies.

(V) Strengthen the prevention and control of heavy metal pollution.

Implement total control of heavy metals. Focus on key industries and key areas such as heavy non-ferrous metal mining, smelting, and electroplating, adhere to strict control of increases and reductions in stock, and continue to promote pollution prevention and control of key heavy metals such as cadmium, mercury, arsenic, lead, chromium, and thallium. Strictly control environmental access for key industries involving heavy metals, and implement

The principle of "equal replacement" and "reduction replacement" will be followed for the emission of key heavy metal pollutants. The efforts to improve the production process of enterprises in non-ferrous metals, electroplating and other industries will be strengthened, and pollution control projects such as the transformation of heavy metal special emission limits will be actively promoted to continuously reduce the emission of heavy metal pollutants. By 2025,

In 2018, the emissions of key heavy metal pollutants in key industries decreased by 5%.

Strengthen the control of mine water pollution. Comprehensively apply satellite remote sensing, drones and on-site surveys to investigate, survey and classify the pollution of heavy metals in mine water throughout the province, find out the bottom line, formulate rectification work plans according to local conditions, and promote pollution control.

Pilot demonstrations for mine water control and management have been carried out in the following areas: the Xinfeng Mine, the Loudi Shaodong Coal Mining Area, the Anhua Taojiang Heshan Coal Mining Area, and the Chenzhou Nonferrous Metal Mining Area.

Strengthen the comprehensive management of tailings ponds. Conduct a comprehensive investigation of tailings ponds, promote the management of tailings ponds in a graded and classified manner, establish a problem list based on cities and prefectures, clarify the responsible parties, management measures, time limits, etc., and accelerate the implementation of management in accordance with the "one pond, one policy" approach.

New (renovation, expansion) tailings ponds will be built within the scope of 3 km of the river, 1 km of important tributaries and the shoreline of Dongting Lake; a sound environmental early warning and monitoring system for tailings ponds will be established with a focus on tailings ponds upstream of drinking water sources; resource utilization of tailings will be encouraged, and unapproved recovery of tailings is strictly prohibited.

Mine and reservoir safety management minimizes the risk of tailings entering farmland due to dam breach and other accidents, and controls environmental risks in mining areas according to local conditions.

Strengthen the management of polluted sites. For agricultural land contaminated by heavy metals, accurately classify the soil environmental quality of cultivated land and implement it to every piece of farmland; carry out in-depth investigation and rectification of the sources of cadmium and other heavy metal pollution, and fully understand the sources of cadmium and other heavy metal pollution; consolidate and deepen the heavy metal pollution of cultivated land in Changsha, Zhuzhou and Xiangtan.

Establish a heavy metal contaminated land information database, a list of construction land soil pollution risk control and restoration, strengthen the connection with construction land approval, and achieve precise control of construction land.

Strengthen the control of key industries. Optimize the industrial layout and continue to eliminate obsolete production capacity in key heavy metal industries. Implement heavy metal pollution control and improvement actions and increase the transformation of production processes. Implement special emission control measures for particulate matter and key heavy metal pollutants for lead, zinc and copper smelting enterprises around cultivated land.

Limits. Strengthen the investigation and rectification of thallium pollution in nonferrous metals, steel and other industries, strengthen the prevention and control of mercury pollution and the implementation of the domestic implementation tasks of the Minamata Convention on Mercury. Include enterprises in industries involving heavy metals such as cadmium in the list of key pollutant-discharging units, and install automatic pollutant monitoring facilities in all of them by the end of 2025.

Strengthen the monitoring of pollutant emissions and surrounding environmental quality in heavy metal industrial parks and key mining enterprises, and strengthen the investigation of environmental risks. Further promote the comprehensive treatment of pollution from heavy metal enterprises around environmentally sensitive areas such as key rivers, lakes, reservoirs, and water sources.

Column 5: Key projects for the prevention and control of heavy metal pollution
Investigation project on enterprises and sewage outlets involving thallium, antimony, etc.: Investigation and survey of enterprises and sewage outlets involving thallium, antimony, etc. in the province . Mine restoration and comprehensive management of mine water: Risk control and pollution management of mine water in closed and withdrawn coal mines and non-coal mines in the province. Heavy metal pollution control technology research project. (1) Harmless treatment project of waste slag containing beryllium in Shuikoushan. (2) Resource utilization and treatment project of bulk manganese slag. (3) Project for resource utilization and harmless treatment of historical chromium slag and arsenic slag.

5. Strengthening ecosystem protection and restoration

1. Build a provincial ecological security pattern.

Build a strong ecological safety barrier. Strengthen the protection of the important ecological function areas of "one river, one lake, three mountains and four waters", and coordinate the protection and restoration of the mountain, water, forest, field, lake, grassland and sand system. Ensure the ecological functions of flood regulation, water conservation, climate regulation and biodiversity protection in the Yangtze River coastline and Dongting Lake area, and promote

The Yangtze River and its shoreline ecosystem will be restored, and the Dongting Lake and its inner lake wetland ecosystem will be restored. The Yangtze River green ecological corridor will be built to improve the connectivity between rivers and lakes, enhance the stability of the ecosystem and the ecological service functions, and protect the habitats and migratory paths of rare and endangered wild animals and plants such as finless porpoises and migratory birds, as well as regional representative wild animals and plants.

We will give full play to the biodiversity maintenance, water conservation, and water-soil preservation functions of the Wuling-Xuefeng Mountain Area, Nanling Mountain Area, Luoxiao-Mubai Mountain Area, and the four water source areas of Xiangjiang, Yuanjiang, and strengthen the protection of native zonal vegetation and rare native plants and animals, accelerate the construction of regional ecological corridors, and strengthen

Protection of natural forests, construction of public welfare forests and afforestation in the upper reaches of the Xiang, Zi, Yuan and Li rivers and on both sides of the river.

Optimize the ecological protection pattern. Build a natural protection area system with national parks as the main body, nature reserves as the basis, and various types of natural parks as supplements. Accelerate the basic work such as the assessment and adjustment of ecological protection red lines, the integration and optimization of natural protection areas, and the demarcation and standardization of boundaries, and strictly control the natural protection.

Non-ecological activities within the protected areas will be prohibited, residents, farmland and mining rights in the core areas will be steadily withdrawn in an orderly manner, and the construction of national parks will be promoted. By 2025, the ecological protection red line and natural protected areas will be fully optimized, adjusted and demarcated, and the proportion of the ecological protection red line area to the national land area will not decrease.

The proportion of nature reserves to the country's total land area remains stable at around 11%.

Strengthen the construction of ecological corridors. Accelerate the construction of ecological corridors, optimize and form large-scale ecological corridors with Wuling-Xuefeng Mountain Range Ecological Corridor, Luoxiao-Mubai Mountain Range Ecological Corridor, and Nanling Mountain Range Ecological Corridor as the main body, and build the Dongting Lake Ecological Corridor, Xiangjiyuanli Four Rivers Ecological Corridor, and other ecological corridors.

We will build medium-scale ecological corridors with the backbone road network as the vein, and improve the small-scale ecological corridors with the backbone road network as the framework. By 2023, the construction of provincial-level important ecological corridors will have made significant progress. We will simultaneously promote the construction of municipal and county-level ecological corridors, and build a continuous, complete, beautiful, and structurally stable ecological corridor across the province.

A stable and fully functional ecological corridor and biodiversity protection network system.

(ii) Improving ecosystem stability.

Promote wetland protection and restoration. Adhere to natural restoration as the main approach, and strengthen the protection of well-preserved natural wetlands in the "one area, four belts, one network, and multiple points" (Dongting Lake area, Xiangjiyuanli four belts, wetland protection system network, wetland protection and construction project points). Promote the restoration of wetlands in East Dongting Lake, West Dongting Lake, and South

We will strengthen the protection and restoration of internationally important wetlands such as Dongting Lake, nationally important wetlands such as Langqian Lake and Jiangkou Bird Island, and other national wetland parks. We will strengthen the protection of wintering water birds and key species in wetlands, and initially form a wetland ecological security pattern with biodiversity protection as the core. We will strengthen wetland use control and

Use supervision to promote the protection and construction of small wetlands and carry out demonstrations of rational utilization of wetland resources in an orderly manner.

Strengthen the governance of ecologically fragile areas. Strengthen the construction of the Yangtze River shelterbelt system, carry out national afforestation, and implement mountain closures for afforestation and grass cultivation. Strengthen the construction of supporting projects such as slope-to-terrace conversion and small-scale water conservancy and water conservation, carry out ecological relocation demonstrations, and mitigate the impact of human activities on ecologically sensitive and fragile areas.

Focusing on the southern Hunan region, we will promote land erosion control in important river source areas, important water sources and key land erosion prevention and control areas, and promote desertification control in the Wuling Mountain area in western Hunan, the Hengshao Basin in central Hunan, and the Chenzhou and Yongzhou regions in southern Hunan.

Accelerate the construction of green mines. Improve the green mine construction work system with government guidance, departmental collaboration, enterprise construction, third-party evaluation, and social supervision, optimize green mine construction standards, improve supporting incentive policies, strengthen supervision and law enforcement, and build a long-term mechanism for green mine development.

Summarize and promote the pilot experience of green development reform in the mining industry, implement the three-year action plan for green mine construction, and ensure that all production mines in the province meet the green mine standards of Hunan Province, thus basically forming a new pattern of green development of mines that is environmentally friendly, efficient and economical, scientifically managed, and harmonious between mines and land.

Huayuan National Green Mining Development Demonstration Zone will be built to give full play to its demonstration and leading role. A number of provincial demonstration mines will be recommended for inclusion in the national green mine list. The ecological restoration of abandoned mines left over from history will be promoted.

Promote the protection and restoration of urban ecosystems. Continue to promote the construction of garden cities and forest cities, enhance the quality of urban charm, promote the construction of "park green space 5 to 15 minutes service circle" projects, comprehensively promote the construction of urban greenways, connect urban parks, green central parks, suburban parks, wetlands

We will build green space systems such as parks and forest parks, and steadily increase the green space rate of urban built-up areas in the province. We will strengthen the protection of natural mountains, river and lake systems, and famous ancient trees, and increase the naturalization rate of urban water bodies and coastlines. We will strengthen the protection of the Changsha-Zhuzhou-Xiangtan Green Heart and improve the ecological quality of the Changsha-Zhuzhou-Xiangtan metropolitan area. We will promote the construction of sponge cities,

Implement comprehensive utilization of urban runoff, urban rivers and artificial wetland water purification projects downstream of sewage outlets according to local conditions to enhance the self-purification capacity of urban water.

(3) Strengthen biodiversity protection.

Strengthen the foundation of biodiversity protection. Promote the revision of biodiversity protection strategies and action plans, compile and publish the "Hunan Province Biodiversity White Paper", and implement major biodiversity protection projects. Further improve the biodiversity supervision system and incorporate biodiversity indicators into the ecological

The quality monitoring, evaluation and effectiveness assessment system will be established to promote the biodiversity survey, observation and assessment of the priority areas for biodiversity conservation in Wuling Mountains, Nanling Mountains and Dongting Lake, establish a basic biodiversity database, strengthen the integrated analysis of basic biodiversity data, information sharing and comprehensive application.

Strengthen the investigation, monitoring and evaluation of rare and endangered species, and master the basic information of species' historical distribution areas, actual distribution areas, population size, population structure, habitat and habitat. Strengthen the supervision of wildlife protection, completely prohibit the illegal trade of wildlife, and eliminate the bad habit of indiscriminate consumption of wildlife.

Strengthen publicity and education on biodiversity protection and biosafety.

Strengthen the protection of flagship endemic species. Strengthen the protection of species of international concern, national key protected species, rare species endemic to Hunan Province, key species and their habitats. Improve the in situ and ex situ protection network for key species, compile and publish the Red List of Rare and Endangered Species in Hunan Province,

We will continue to promote the standardization and regularization of nature reserves, ecological corridors and other protected areas, and improve the system of ex situ conservation of zoos and botanical gardens, aquariums, germplasm resource banks (nurseries), microbial strain collection centers, and rare and endangered plants. By 2025, the province will have 1,000 national key protected wild animals and plants.

The protection rate is maintained at above 85%, and typical ecosystems are effectively protected.

Promote the protection and management of genetic resources. Promote the establishment of a management system for access to and benefit sharing of biological genetic resources, and strengthen the management and supervision of foreign cooperative research and utilization. Carry out a census of germplasm resources, conduct a systematic investigation of rare, endangered, unique, special germplasm resources and local variety resources, and seize the

Rescue collection of ancient local varieties, long-standing bred varieties and other rare and endangered resources. Carry out rescue collection and protection of rare and endangered wild plant resources in Hunan and research on the discovery of excellent genes, and carry out pilot projects on the benefit sharing of biological genetic resources and related traditional knowledge.

Strengthen biosafety management. Strengthen the control of invasive alien species, continue to carry out investigations, monitoring and early warning of invasive alien species, strengthen research on invasion mechanisms, diffusion pathways, response measures and development and utilization pathways, actively prevent and control the invasion of alien species, eliminate the threat of invasive species, and promote the development of

Carry out pilot projects for the removal of harmful alien species. Continue to improve the monitoring, prevention and control management level of alien invasive species and wildlife epidemic sources and diseases, establish and improve inter-departmental coordination mechanisms, strengthen technical system research, and promote the demonstration and application of prevention and control technologies. Strengthen environmental safety supervision of genetically modified organisms.

(IV) Strengthen natural ecological supervision and management.

Establish a natural ecological supervision system. Promote local legislation in nature reserves, establish and improve natural ecological protection supervision mechanisms and standards, and continuously improve the legalization, institutionalization, and standardization of supervision. Strengthen the supervision of natural ecological protection through off-site supervision, drone supervision, satellite remote sensing and other application technologies.

Supervision of behaviors that damage wetlands, forests, grasslands, and natural coastlines, as well as wetland ecological environment protection, desertification prevention, and coastline protection and restoration. Regularly evaluate the effectiveness of nature reserves and ecological protection red lines, promote performance appraisal and supervision and accountability, and increase supervision of those who occupy ecological space and damage.

The severity of the penalties for behaviors that harm important ecosystems.

Strengthen the supervision of the ecological protection red line. Implement the supervision responsibilities of the ecological protection red line, conduct a background survey of the ecological environment and human activities within the ecological protection red line, and determine the baseline level of ecological functions of the ecological protection red line. Carry out a pilot project to supervise the ecological destruction of the ecological protection red line, and establish a supervision working mechanism.

system, accelerate the construction of the ecological protection red line supervision platform, and strengthen monitoring, assessment and early warning.

Promote supervision of nature reserves. Organize and carry out the "Green Shield" nature reserve strengthening supervision, establish and improve the ecological and environmental problem ledger of nature reserves, and conduct on-site verification of key issues. Strengthen the supervision of the establishment, promotion (downgrading), adjustment, integration and withdrawal of nature reserves, and conduct supervision of the areas involved in the adjustment.

Promote the rectification of outstanding ecological and environmental problems in nature reserves, strictly implement the rectification and cancellation system, and urge key issues to be investigated and dealt with in accordance with the law and rectified thoroughly. Strengthen inter-departmental coordination and linkage, incorporate key issues into the provincial ecological and environmental protection supervision and coordinate the rectification, and consolidate

Effectiveness of problem rectification.

Strengthen the monitoring and evaluation of ecological quality. Explore the implementation of ecological quality monitoring and evaluation work in the province and key ecological functional areas, and fully grasp the changes and trends of ecological conditions; carry out remote sensing monitoring and evaluation of key ecological functional areas and ecological civilization demonstration areas in counties. Carry out ecological and environmental protection of nature reserves.

Conduct environmental protection effectiveness evaluation to timely grasp the management and protection effectiveness of nature reserves. Organize and carry out the implementation effectiveness evaluation of ecological protection and restoration projects, and monitor the changes in ecological quality and environmental quality throughout the process. Explore the implementation of ecological impact monitoring and evaluation of natural resource development and utilization activities to provide ecological protection and restoration services.

Provide support for supervision and law enforcement. Strengthen the application of monitoring and evaluation results, use monitoring and evaluation results as an important basis for ecological protection compensation, transfer payments for key ecological function areas, and allocation of special funds for ecological protection, restoration and management, and include them in relevant assessments of ecological civilization construction.

(V) Promote the creation of ecological civilization models.

Continue to promote the creation work. Guided by the joint efforts of the ministry and the province to strengthen the pilot work of ecological civilization demonstration creation, promote the creation of national environmental protection model cities, ecological civilization demonstration cities (counties), "Green water and green mountains are gold and silver mountains" practice and innovation bases, and provincial ecological civilization demonstration cities (counties).

Promote the establishment of a provincial ecological civilization demonstration work coordination mechanism, guide each creation area to form a creation work pattern in which each performs its duties and promotes the work in a coordinated manner; explore the mechanism for applying the results of ecological civilization demonstration creation, increase support for national environmental protection model cities, ecological civilization construction demonstration areas and "green water and green mountains are

The incentives for the "Golden Mountain and Silver Mountain" practice and innovation bases will be included in the scope of the provincial government's real work supervision and incentives; continue to strengthen the publicity and promotion of successful experiences and typical models of ecological and cultural demonstration creation to enhance the influence of demonstration creation. Strengthen classified guidance, implement hierarchical management, and promote important projects in different regions.

The work of creating ecological civilization models is carried out at the points.

Column 6 Key projects for natural ecological protection and restoration
Ecological security barrier protection and restoration project. Carry out ecological protection and restoration of Wuling-Xuefeng Mountain, Nanling Mountain, Mufu-Luoxiao Mountain, and ecological protection and restoration of the Hunan section of the Yangtze River coastline. Changsha-Zhuzhou- Xiangtan Ecological Green Heart Protection and Restoration Project. With the goal of creating an urban green heart with international quality, promote ecological culture, and systematically improve the ecological environment of Changsha-Zhuzhou-Xiangtan. The ecological green heart area has ecological service functions such as biodiversity protection, water conservation, soil conservation, water source protection, and climate regulation. Ecologically fragile area governance. Comprehensive desertification control projects, comprehensive control of soil erosion in small watersheds. Forest land construction. Natural forest (public welfare forest) protection and restoration projects, key shelterbelt construction projects.

Hunan Provincial People's Government Portal Green Mine Construction. Promote the ecological restoration of abandoned mines left over from history, focus on solving the ecological damage of open-pit mines left over from history, and strengthen mining. Comprehensively improve the slopes, reshape the terrain, rebuild ecological vegetation, restore the ecological environment of the mining area, and implement green mine construction.

Biodiversity conservation project. Carry out a comprehensive survey of the biodiversity conservation priority areas and establish a wild resource base for rare and endangered wild animals and plants in the province. Monitoring system to promote the protection of migratory birds, herbivores, big cats and their habitats.

Monitoring and evaluation of ecological quality. Remote sensing monitoring and evaluation of key ecological function areas and ecological civilization demonstration areas in counties, and ecological environment protection in nature reserves. Evaluation of ecological quality of the whole province and ecological protection red line.

VI. Prevent and resolve ecological and environmental risks

1. Strengthen the management and control of hazardous waste.

Strengthen the supervision of the entire process of hazardous waste. Adhere to the principle of "strict outside the province and active within the province", establish a long-term mechanism for the environmental management of hazardous waste, improve the environmental management system of hazardous waste, and promote the hierarchical and classified management system. Under the premise that environmental risks are controllable, carry out "point-to-point" targeted utilization of hazardous waste. The pilot project of hazardous waste exemption management shall be launched; the level of hazardous waste management informatization shall be improved, and a complete hazardous waste informatization supervision system that can be "located, shared, and traced" shall be established to realize the "one network" of hazardous waste informatization management in the province; standardized management of hazardous waste shall be promoted, and illegal transfer of hazardous waste shall be severely cracked down. Illegal activities such as the transfer, dumping, disposal and unlicensed operation of hazardous waste.

Strictly control the environmental access of hazardous waste projects. Strictly control the construction of new (expanded) hazardous waste comprehensive utilization projects with excess comprehensive utilization capacity in the province and with raw materials from other provinces as the main source; no new organic hazardous waste thermal (cracking) decomposition treatment projects will be built; the quantity, type, property, storage and management of hazardous wastes will be strictly controlled. Strictly approve construction projects with unclear facilities, no reasonable utilization and disposal plan, no disposal route for secondary solid waste, and no environmental risk prevention measures; promote hazardous waste generating units to optimize process, equipment and raw material selection to reduce the generation of hazardous waste at the source. Explore the inclusion of hazardous waste in pollution discharge permits Scope of certificate management.

Coordinate the layout of hazardous waste disposal facilities. Fully grasp the number of hazardous wastes in the province; gradually promote large and medium-sized hazardous waste production enterprises (more than 5,000 tons/year) to build supporting self-use disposal facilities; encourage chemical, smelting and other industrial parks to build supporting centralized storage and pre-treatment facilities for hazardous wastes; Management and disposal facilities will be established to promote the rational layout of thallium-related waste utilization and disposal facilities in key areas; a number of hazardous waste treatment facilities will be built with centralized incineration and landfill as the main method and cement kilns as the auxiliary method; and a hazardous waste utilization and disposal system with overall matching capacity will be established within the province.

Improve the hazardous waste collection, transportation and transfer system. Carry out pilot projects for the centralized collection and storage of hazardous waste; promote the implementation of the extended producer responsibility system, encourage production and operation units to establish professional service teams and collection sites; encourage the reasonable construction of hazardous waste storage facilities in accordance with the law based on the actual situation of the local area. measures; promote the professionalization, scale and park-based development of hazardous waste classification collection; explore the establishment of industrial parks or industry hazardous waste collection platforms, and improve the transportation capacity of hazardous waste collection in small and micro-enterprise industrial parks, scientific research institutions, etc.; standardize the lead-acid battery and waste mineral oil recycling network system; Strictly control the inter-provincial transfer of hazardous waste, promote the establishment of a black (white) list system for the inter-provincial transfer of hazardous waste, and establish a regional joint prevention and control mechanism for hazardous waste environmental risks.

Strengthen the medical waste disposal capacity. Accelerate the construction of Xiangtan City's centralized medical waste disposal center; encourage existing medical waste disposal companies to upgrade technology and improve quality and expand capacity; strengthen the management of medical waste collection, storage, and disposal, and strengthen the operation and supervision of existing facilities. Strengthen the construction of medical waste coordinated emergency disposal mechanism through the centralized disposal center for hazardous waste and the coordinated treatment of cement kilns. Improve the disposal material reserve system to ensure the emergency disposal capacity of medical waste in major epidemics.

Promote the comprehensive utilization of general industrial solid waste. Encourage local people's governments at or above the county level to coordinate or jointly plan and build general industrial solid waste centralized disposal facilities, support the research and development and application of new technologies, new equipment, and new products for resource utilization; make full use of industrial solid waste under controllable environmental risks. Industrial kilns, cement kilns and other facilities to dispose of large-scale industrial solid wastes such as mining tailings, fly ash, slag, smelting waste residue, desulfurized gypsum, etc.; build an industrial solid waste comprehensive utilization system with cement, building materials, metallurgy and other industries as the core; promote the comprehensive utilization of industrial solid waste resources demonstration base (parks), demonstration enterprises and demonstration projects will be built. By 2025, the comprehensive utilization rate of general industrial solid waste resources in the province will reach 80%.

(ii) Strengthen environmental management of chemicals.

Strengthen the risk control of new pollutants. Strictly implement the requirements of international conventions on chemicals and the environment, focus on persistent organic pollutants, endocrine disruptors, antibiotics and microplastics, and continuously improve the environmental risk assessment and control capabilities of new pollutants; cooperate to promote the foundation of chemical substance risk assessment. Database construction.

Strengthen the supervision of the disposal of waste hazardous chemicals. Continue to carry out standardized environmental management of waste hazardous chemicals; urge enterprises to fulfill their principal responsibilities and carry out solid waste identification and hazardous waste identification for intermediate products (products), by-products (products) and other materials with unclear qualitative characteristics.

(3) Strengthen nuclear and radiation safety supervision.

Ensure the safety of the radiation environment in the use of nuclear technology. Continue to regularly carry out special actions for safety inspection of radioactive sources, carry out the standardization of radiation safety management in nuclear technology utilization units, build an online real-time monitoring platform for high-risk mobile radioactive sources, and fully realize real-time monitoring of high-risk mobile radioactive sources. Strengthen the safety management of waste radioactive sources, evaluate and upgrade waste repositories, regularly clear repositories, ensure that 100% of waste radioactive sources found are safely stored, strengthen the clean clearance management of radioactive waste using nuclear technology. Strengthen radiation safety supervision of scrap metal recycling and smelting enterprises, and urge enterprises to conduct radiation monitoring independently.

Strengthen electromagnetic radiation management. Strengthen the legality review of electromagnetic projects. Build a provincial electromagnetic radiation monitoring system, select typical electromagnetic radiation facilities for supervisory monitoring, and set up provincial electromagnetic environment quality monitoring points in key areas or densely populated areas. Make every effort to promote the control of electromagnetic radiation (noise We will deploy online automatic stations for monitoring the environment (such as acoustic) to realize the public release of real-time data information. We will solve the problems of public communication and public opinion resolution, carry out special investigation and rectification actions in the fields of mobile base stations and power transmission and transformation projects, and do a good job in popularizing electromagnetic science.

Accelerate the resolution of historical problems. Carry out environmental pollution control for radioactive mining enterprises. Improve the environmental safety supervision capacity of associated mines, establish and dynamically update the list of enterprises that supervise the radiation environment safety of the development and utilization of associated radioactive mines, and urge all operating enterprises in the list to carry out radiation environment safety supervision. Conduct radioactive contamination surveys at sites of closed enterprises and their surroundings, and promote land reuse at closed associated mining sites. Promote standardized management of radioactive solid waste from associated mining, standardize temporary storage of solid waste, and control reuse. Conduct radioactive solid waste management surveys at sites of closed associated mining sites, and Conduct feasibility research on centralized waste disposal and promote the construction of radioactive solid waste disposal sites.

Strengthen the basic guarantee of nuclear and radiation safety. Efficiently operate the Hunan Provincial Nuclear Safety Coordination Mechanism, strengthen administrative approval and in-process and post-process supervision, build a provincial radiation environment supervision big data platform, and promote the city and prefecture ecological and environmental departments to be equipped with full-time radiation safety supervision personnel. Improve the emergency command and control system at all levels. We have established a comprehensive emergency response system, established and improved the emergency notification and liaison system, carried out the formulation of nuclear emergency plans, built an emergency command platform that deeply integrates peacetime and wartime operations, strengthened the allocation of radiation emergency supplies, established a radiation emergency rescue material reserve system, and completed a round of comprehensive radiation accident emergency drills in all provinces and cities. Strengthen the integration of the radiation environment monitoring network and the conventional environment monitoring network, and actively promote the improvement of the radiation monitoring capabilities of the provincial radiation station and the four branch centers in Changsha, Hengyang, Yiyang and Huaihua.

(IV) Strengthen emergency preparedness for environmental risks.

Strengthen ecological and environmental protection monitoring. Strengthen the construction of ecological and environmental protection monitoring platforms and ecological and environmental monitoring centers, build an "ecological environment + power big data" smart supervision platform, and realize the integrated monitoring of "environmental quality, pollution source emissions, smart electricity, and pollution source videos". Protect daily supervision and offline monitoring, improve the environmental risk early warning and control system, promote risk prevention and control in key river basins and important water sources, and the construction of early warning systems for chemical industrial parks and toxic and hazardous gases; establish an ecological and environmental risk hazard investigation system and a database of major ecological and environmental risk sources, and implement dynamic Tracking, monitoring and management; establish ecological and environmental risk supervisors to promptly verify issues reported by the public, public opinion and other channels, establish problem lists and rectification lists, and eliminate environmental risks.

Strengthen emergency response to emergencies. Strengthen the response to ecological and environmental monitoring information, and quickly issue warnings, push notifications, verify, and handle data anomalies, major risk hazards, and major public opinions found in ecological and environmental monitoring to prevent the expansion of pollution, the outbreak of risks, and the escalation of incidents. Focus on the prevention and mitigation of ecological and environmental incidents, tailings ponds, mining and smelting enterprises, etc., improve the risk prevention and mitigation and emergency preparedness responsibility system for sudden ecological and environmental incidents, and strictly implement the main responsibility of enterprises. Improve the ability to properly handle the incident, improve the emergency response and disposal mechanism for sudden ecological and environmental incidents, and enhance emergency response Standardize the level of disposal. Promote post-event damage assessment, ecological and environmental damage compensation, environmental restoration, and environmental public interest litigation.

Improve the level of emergency response and guarantee. Urge enterprises to formulate emergency plans for sudden environmental incidents, and achieve full coverage of electronic filing for hazardous and heavy enterprises. By the end of 2022, governments at the county level and above will conduct risk assessments of sudden environmental incidents in their administrative areas and complete the revision of emergency plans, and promote key river basins. Special emergency plans for upstream and downstream water environment incidents; establish a joint prevention and control mechanism for upstream and downstream water pollution incidents; promote the experience of "Nanyang Practice" to achieve full coverage of "one river, one policy, one map" for environmental emergency response in key rivers; promote the construction of a "government + enterprise" environmental emergency material guarantee system, and promote the construction of Establish regional environmental emergency material reserves; improve the multi-level environmental emergency expert management system, revise the expert management method, and clarify the expert intervention mechanism; carry out rotation training for grassroots environmental emergency personnel in a classified and graded manner, strengthen emergency tracing, disposal, monitoring equipment and emergency vehicle configuration, and improve environmental Build an emergency command platform, strengthen emergency training and emergency drills, improve the environmental emergency linkage mechanism, and enhance the ability of coordinated disposal.

Strengthen ecological and environmental health management. Organize and implement actions to improve residents' ecological environment and health literacy, form an expert team, carry out popular science activities on ecological environment and health literacy, build a base for improving the ecological environment and health literacy of young people, and establish provincial-level representative residents' ecological environment and health literacy according to actual conditions. Establish an ecological environment and health literacy monitoring network and carry out monitoring work. Promote the identification and investigation of ecological environment health risks, encourage regional ecological environment and health surveys and assessments, and strengthen environmental and health science research in the fields of biological safety and indoor environmental health.

Column 7 Key projects for environmental risk prevention
Solid waste treatment and utilization technology research project. (1) Waste incineration fly ash water washing dechlorination pretreatment project, with a single treatment scale of 30,000 tons/year. (2) Steel Solid waste disposal project for iron and steel enterprises kilns. (3) Urban mineral pollution-free recycling project, with an additional 10 million sets of household appliances dismantling annual processing capacity. Emergency infrastructure support project. (1) Regional environmental emergency material reserve construction project. (2) Emergency equipment and vehicle filling and replenishment project. Urban radioactive waste storage waste source cleaning and warehouse quality improvement project. (1) Carry out radioactive waste source cleaning and transportation, with an estimated number of cleaned items of 2,100. Radioactive sources. (2) Improve the quality of the pit bottom and warehouse of Hunan Province's urban radioactive waste storage No. 1 and No. 2 according to the new requirements. Resource recycling projects. (1) Park recycling transformation. (2) Construction of a national-level large-scale solid waste comprehensive utilization base. Domestic waste treatment projects. (1) Construction of domestic waste incineration power generation facilities at or above the county level. (2) Domestic waste classification front-end projects. (3) Construction of kitchen waste treatment facilities at or above the county level.

VII. Improve the ecological environment governance system

1. Improve the ecological environment governance responsibility system.

Improve the leadership responsibility system. Strictly implement the shared responsibility of the Party and the government, dual responsibility for one post, and accountability for dereliction of duty in ecological and environmental governance. Implement the "Hunan Province Ecological and Environmental Protection Work Responsibility Regulations" and the "Hunan Province Major Ecological and Environmental Issues (Incidents) Accountability Accountability Measures", and hold people accountable in accordance with regulations, discipline, and laws. We will thoroughly implement the system of outgoing audits of natural resource assets of leading cadres and lifelong accountability for ecological and environmental damage. We will scientifically carry out evaluation and assessment, reasonably set binding and expected targets, and incorporate them into special plans such as national economic and social development plans at all levels, national land space planning, and ecological and environmental protection. Improve the comprehensive performance evaluation and assessment system for high-quality development in Hunan Province, and use the assessment results as an important basis for comprehensive evaluation, assessment, rewards, punishments and appointments and removals of leadership teams and leading cadres at all levels.

Improve the corporate responsibility system. Strengthen the pollution control responsibility of enterprises, urge enterprises to effectively increase pollution control investment, improve clean production level, strictly implement environmental impact assessment approval requirements, reduce pollution emissions, strengthen pollution control, and do a good job in ecological restoration. Strengthen the legal compliance responsibility of enterprises, guide enterprises to firmly establish Establish new development concepts, strengthen awareness of law-abiding, consciously strengthen daily management, and guide enterprises to consciously abide by the law; strengthen daily supervision and law enforcement, maintain a high-pressure situation to crack down on illegal pollution discharge, and establish a positive incentive mechanism and a severe punishment mechanism for violations. Strengthen corporate social responsibility and guide enterprises to carry out self-discipline supervision. Measure and actively disclose to the public and accept social supervision; urge enterprises to install and improve online monitoring facilities for pollution sources and strengthen the construction of online monitoring platforms; establish and improve the corporate credit evaluation system, deepen the reform of the environmental information disclosure system in accordance with the law, and promote the mandatory disclosure of environmental information by enterprises in accordance with the law.

Improve the citizen action system. Give full play to the role of the "12369" complaint hotline, use columns such as "Sansiang Ecology" to proactively expose ecological and environmental violations. Improve the news release system for ecological and environmental protection, and proactively respond to major social concerns. Establish a reward system for ecological and environmental violations in Hunan Province. Establish a reporting system and expand channels for public participation and supervision. Strengthen the management and guidance of social organizations involved in ecological and environmental protection, guide and support the construction of ecological and environmental protection volunteer teams, actively expand ecological and environmental protection business training, and improve volunteer service capabilities. Incorporate ecological and environmental protection into the leadership of the Party and government The content of cadre education and training will be included in the corresponding courses offered in party schools (administrative colleges) at all levels. We will promote the establishment of environmental science popularization bases, strive to build about 10 provincial ecological and environmental science popularization bases, and strive to create 1-2 national ecological and environmental science popularization bases.

(ii) Improve the legal and regulatory system for ecological and environmental governance.

Improve the system of laws and regulations. Promote local legislation such as the Hunan Provincial Water Pollution Prevention and Control Regulations and the Hunan Provincial Water Resources Management Regulations, and improve the ecological environment damage compensation management system. Encourage cities and prefectures to make efforts in air pollution prevention and control, natural ecological protection, livestock and poultry breeding pollution prevention and control, and straw comprehensive utilization. Local regulations will be introduced in areas such as the protection of the ecological environment. Strict enforcement of ecological and environmental protection laws will be implemented. Those who cause ecological and environmental damage will be held liable for compensation in accordance with the law, and those who constitute a crime will be held criminally liable in accordance with the law.

Improve environmental protection standards. Organize the formulation and revision of standards for environmental quality and pollutant emissions, and accelerate the formulation of standards and methods for the environmental quality of non-drinking surface water sources and the evaluation of the effects of heavy metal pollution remediation in agricultural soil. Promote ecological and environmental protection planning, environmental We will improve the coordination between environmental protection standards and industrial policies, and improve the information feedback and evaluation mechanism for standard implementation. We will encourage the development of green certification systems involving various types of environmental governance.

Improve the ecological and environmental protection investment mechanism. Establish an incentive mechanism for the creation of ecological civilization demonstration projects, broaden the fundraising channels, operate the provincial soil pollution prevention and control fund, subscribe to the national green development fund and strive for fund support. Improve the ecological compensation mechanism for the entire "one lake and four rivers" basin, and actively promote the construction of Establish a cross-provincial horizontal ecological protection compensation mechanism, carry out cross-provincial river basin ecological protection compensation with neighboring provinces such as Chongqing and Jiangxi Province, and promote the ecological compensation work of the Nanshan National Park system pilot. Promote the development of ecological environmental pollution liability insurance, and study the establishment of a compulsory ecological environmental pollution liability insurance system. Deepen the pilot work of emission rights trading, study and improve emission rights trading policies, and promote the reform of emission rights trading mortgage financing. Strengthen the special guidance on the issuance of green bonds by enterprises and regions. Improve the sewage and garbage treatment charges and differentiated electricity price policies.

Improve the ecological environment damage compensation system. Strengthen the implementation and supervision of ecological environment restoration and damage compensation, improve the administrative law enforcement and judicial coordination mechanism for ecological environment damage, increase the cost of illegal acts that damage the ecological environment, and establish and improve the ecological environment damage identification and assessment methods and implementation mechanisms.

(iii) Improve the market system for ecological environment governance.

Build a standardized and open market. Increase the reform of "delegating power, deregulating and improving services" in the field of ecological environment, break down regional and industry barriers in the field of ecological environment public resource transactions, treat all types of enterprises equally, and guide capital to participate in ecological environment governance investment, construction, and operation. Standardize the bidding market order for ecological environment governance projects and projects supported by special ecological environment funds, and accelerate the formation of an open, transparent, standardized and orderly ecological environment governance market environment.

Strengthen the ecological and environmental protection industry. Formulate a three-year action plan to strengthen the environmental protection industry chain, support key environmental protection enterprises, expand and strengthen a number of leading energy-saving and environmental protection enterprises, cultivate leading environmental protection management enterprises and specialized "small giant" enterprises in a layered manner, and build a number of professional environmental protection parks. Carry out platform creation We will carry out activities such as construction, production and demand matching, product promotion, and brand building, recommend a number of new technologies, new products, and demonstration projects in the fields of environmental protection equipment manufacturing, heavy metal pollution prevention and control, comprehensive resource utilization, ecological restoration, and water quality online monitoring technology and equipment, and support high-quality environmental protection enterprises to take on offshore projects. We will "go global" by developing environmental protection projects abroad and conducting industry mergers and acquisitions, and participate in the construction of the "Belt and Road". We will continue to increase our independent innovation and scientific and technological research and development efforts, support enterprises, universities, and research institutes to establish environmental governance technology innovation and entrepreneurship bases, and actively promote the first set of environmental protection facilities. By 2025, We will strive to maintain an average annual growth rate of 10%-15% in the total output value of the province's environmental protection industry, cultivate more than 50 environmental protection enterprises with core technologies and competitiveness, and ensure that environmental protection scientific research capabilities and technologies generally reach the domestic advanced level.

Innovate the environmental governance model. Accelerate the implementation of third-party governance of industrial park environmental pollution according to local conditions, establish and improve mechanisms such as "polluter pays + third-party governance", and by the end of 2022, provincial-level and above industrial parks will basically achieve full coverage of third-party governance of environmental pollution. Actively create ecological and environmental protection in small towns. We will launch a national pilot project for comprehensive environmental governance trusteeship services, strengthen system governance, and implement pay-per-performance. We will strengthen the information coordination of polluted land and the prevention and control of environmental risks in safe use, and encourage the adoption of the "environmental restoration + development and construction" model for industrial polluted land.

Accelerate the construction of environmental credit. Promote the construction of environmental credit in industrial parks and enterprises and institutions, improve the enterprise environmental credit evaluation and result release mechanism, promote the reform of the "dishonest list" management system, study and revise the management measures for the ecological and environmental protection dishonest list, carry out credit constraints and joint Punishment. Upgrade the "Hunan Enterprise Environmental Credit Rating System" and gradually achieve interconnection with other credit platforms in the province.

(IV) Improve the mechanism for realizing the value of ecological products.

Establish an ecological product value accounting mechanism. Using natural resource rights registration and ecological product information census as basic means, establish an open and shared ecological product information cloud platform. Establish an ecological product value evaluation mechanism and standardize ecological product value accounting. Promote the results of ecological product value accounting. Application in government decision-making and performance evaluation to quantify the value of ecological products.

Build an ecological product business transaction system. Promote the construction of ecological product transaction centers, standardize platform management, give full play to the resources and channel advantages of e-commerce platforms, and promote more high-quality ecological products to be traded through convenient channels and methods. Under the premise of strictly protecting the ecological environment, we encourage the adoption of various We will establish a new model and path to promote the value of ecological products in a scientific and reasonable manner. We will encourage the creation of distinctive regional brands of ecological products, include various types of ecological products in the brand range, strengthen brand cultivation and protection, and increase the premium of ecological products.

Improve the compensation mechanism for ecological product protection. Improve the vertical ecological protection compensation system and improve the allocation mechanism for transfer payments to key ecological functional areas. Further expand the horizontal ecological protection compensation mechanism and explore the extension of horizontal compensation to areas such as regional air quality and hazardous solid waste disposal. Explore compensation models for off-site development, establish cooperative parks between ecological product suppliers and beneficiaries, and improve benefit distribution and risk sharing mechanisms.

Carry out pilot projects on the value realization mechanism of ecological products. In conjunction with the creation of national ecological civilization demonstration cities (counties) and "Green waters and green mountains are gold and silver mountains" practice and innovation bases, carry out pilot projects on the value realization mechanism of ecological products, focusing on ecological product value accounting, precise matching of supply and demand, and sustainable development. Carry out practical exploration in business development, protection compensation, evaluation and assessment, etc. Summarize successful experiences, strengthen publicity and promotion, and create a number of demonstration bases for the value realization mechanism of ecological products.

(V) Strengthen the capacity building for ecological and environmental governance.

Strengthen the construction of the ecological environment team. Consolidate and deepen the results of the reform of the vertical management system of the ecological environment, comprehensively strengthen the construction of the cadre team of the ecological environment system, always put the political construction of the party in the first place, and use party building as a guide to lead the team, lead the team, and promote work. Unwavingly promote comprehensive and strict governance The Party adheres to the general tone of strictness and perseverence in rectifying the style and discipline. Strictly implement the eight central regulations and their implementation rules and the provincial party committee's implementation measures, and resolutely rectify the problems of formalism and bureaucracy. Continue to carry out special rectification of party style and clean government and special rectification of bureaucracy and formalism to promote party building.

We will increase the training and exercise of young cadres, strengthen talent training and the political, ideological, organizational, work style and discipline construction of the cadre team, carry out the "big rotation training" of the provincial ecological and environmental protection system, implement the "three one hundred" talent plan of the provincial ecological and environmental protection system, and improve political

Literacy and business skills.

Strengthen the capacity building of ecological and environmental supervision. Implement the "Hunan Province Ecological and Environmental Protection Supervision Work Implementation Measures", formulate a provincial ecological and environmental protection supervision plan, and earnestly carry out routine supervision, supervision "look back", special supervision, and daily supervision. Expand the scope of supervision, respond to climate change,

The implementation of major decisions and arrangements such as biodiversity protection will be included in the scope of supervision. We will improve our ability to identify and solve problems, establish and improve the "1+X" long-term mechanism for supervision work, expand the sources of information on prominent ecological and environmental problems, and form a joint force for supervision and rectification work. We will accelerate the promotion of supervision informatization and complete

The Phase II construction of the Hunan Provincial Ecological and Environmental Protection Supervision Management Information Platform will establish a unified database of outstanding ecological and environmental problems in the province to achieve real-time supervision, unified supervision, unified dispatch and feedback on completion. Promote the rectification of outstanding ecological and environmental problems, strengthen the dispatch of rectification and supervision,

Take measures such as early warning, written notification, notification, supervision, listing for supervision, interviews, exposure, regional approval restrictions, investigation and accountability to increase the promotion efforts. Establish and improve the rectification and cancellation system for prominent ecological and environmental problems, formulate specific measures to strictly prevent false rectification, superficial rectification, and pretend rectification, as well as petitions.

The standards for acceptance and cancellation of cases shall be strictly enforced, and the provincial-level verification process and requirements shall be refined to ensure the quality of rectification and cancellation.

Strengthen the construction of ecological and environmental monitoring capabilities. Comprehensively deepen the reform and innovation of ecological and environmental monitoring, further improve the ecological and environmental monitoring pattern, improve the ecological and environmental monitoring work responsibility system, and improve the quality of monitoring data. Optimize the ecological and environmental monitoring network and establish a network covering the atmosphere, surface water, and underground

Establish a provincial environmental quality monitoring network for environmental factors such as water, drinking water, soil, noise, radiation, and greenhouse gases; promote key industrial parks and industrial clusters to establish a volatile organic compound and particulate matter monitoring system; strengthen the monitoring of urban and rural black and odorous water bodies, sewage treatment plants, garbage disposal sites, and river discharges;

Monitoring of pollutants and rural non-point sources; setting up more ecological environment quality monitoring stations to promote ecological environment quality monitoring and evaluation. Improving the level of ecological environment monitoring services, improving the ecological environment quality evaluation system, scientifically and objectively reflecting the status and changes of environmental quality; strengthening environmental risk prediction and prevention

Strengthen emergency monitoring of emergencies, and establish a regional joint response mechanism; strengthen the analysis of pollution sources, causes and transmission, and improve the ability to analyze the correlation between ecological environment quality and pollution emissions. Improve the modernization level of ecological environment monitoring, and comprehensively improve the automation, standardization, and

The level of informatization should be strengthened; the county-level ecological and environmental monitoring capacity should be strengthened, local standards for county-level ecological and environmental monitoring capacity should be developed, and personnel and equipment should be gradually supplemented according to regulations. By 2025, county-level ecological and environmental monitoring agencies should be able to effectively carry out law enforcement monitoring and emergency response within their administrative areas.

Ability to monitor.

Strengthen the capacity building of ecological and environmental law enforcement. Deepen the reform of comprehensive administrative and law enforcement for ecological and environmental protection, and promote the inclusion of ecological and environmental law enforcement in the administrative system to ensure its protection. In accordance with national requirements, comprehensively promote the standardization of law enforcement teams, complete the standardization of uniforms and uniforms for environmental law enforcement personnel at the provincial, municipal and county levels, and complete the standardization of uniforms and uniforms for environmental law enforcement personnel at the provincial, municipal and county levels.

The distribution of signs will fully realize the uniforms for environmental law enforcement officers; the law enforcement vehicles, personal mobile law enforcement equipment, on-site law enforcement auxiliary equipment, communication and office equipment, information equipment and other law enforcement equipment will be fully equipped. The structure of law enforcement personnel will be optimized. By 2025, the number of law enforcement personnel with college degree or above will reach

More than 90% of the employees have obtained a degree in law or ecological environment related majors, and more than 50% have obtained a degree in law or ecological environment related majors. Improve the standardization of law enforcement, formulate and timely update the "List of Comprehensive Administrative Law Enforcement Items of Hunan Province Ecological Environment", implement "double random, one public", standardize free discretion, and formulate positive supervision and law enforcement.

We will continue to promote the informatization of ecological and environmental law enforcement, improve the ecological and environmental supervision and law enforcement system based on mobile law enforcement, strengthen the construction of online monitoring and electricity monitoring, and use technological means such as drones, drones, UAVs, and satellite remote sensing to innovate non-site supervision methods. By 2025

By 2020, a comprehensive administrative and law enforcement system for ecological and environmental protection will have been basically established, with a sound system, unified uniforms, complete equipment and facilities, and compatible with the cause of ecological and environmental protection.

Strengthen the construction of ecological environment informatization. Promote the construction of ecological environment informatization pilot provinces, improve the digital governance capabilities of the ecological environment, and accelerate the construction of a "big support platform", "big data center" and "big application system" for ecological environment protection jointly built and shared by the province, so as to realize data sharing, system integration and management.

We will strengthen the province's unified planning and project management, establish an information project database, and realize the "one game" of information construction of the province's ecological environment system. The system integrates the environmental protection network, the electronic government extranet and the government cloud, and the safe and reliable

We will use reliable commercial cloud and other network, computing and storage resources to build the ecological environment infrastructure network and ecological environment cloud of Hunan Province. With the provincial ecological environment big data resource center as the main node, we will open up the links with the ecological environment data resource centers of cities and prefectures, gradually establish a distributed storage system, and build a national ecological environment big data resource center.

The province will jointly build and share an "ecological environment big data center" with unified logic. Build a "one map of the ecological environment" to provide common geospatial data applications and services for the provincial, municipal and county levels. Establish a directory of ecological environment protection business components for the province to provide unified support for the development of business applications, and gradually

Build an ecological and environmental business application system with Hunan characteristics. Coordinate the construction of the province's ecological and environmental protection "comprehensive management platform" and "government service platform" to support the integration, deployment and management of various business applications on the two platforms. Promote the construction of regional ecological and environmental informationization based on government, schools, research and enterprises

Cooperate with exchange institutions to jointly carry out information technology research and talent training. Build and improve the ecological environment network security system to provide strong security guarantees for information technology construction.

Strengthen the construction of ecological and environmental scientific research capabilities. Strengthen the construction of ecological and environmental protection scientific and technological innovation system to solve the "bottleneck" ecological and environmental technical problems in our province. Carry out heavy metal pollution control, PM2.5 and ozone coordinated control, "one lake and four waters" total phosphorus control, arsenic alkali slag beryllium slag and other hazardous waste treatment, soil pollution control, etc.

We will conduct research on key and common technologies such as pollution risk control and remediation, intensive management of rural life pollution, and agricultural non-point source pollution control, and strive to achieve domestic leading levels in several core technologies. We will further promote scientific and technological innovation in water safety of "one lake and four waters", scientific and technological innovation in air pollution prevention and control, and scientific and technological innovation in soil restoration technology.

Accelerate the construction of the heavy metal pollution prevention and control center jointly built by the ministry and the province, strengthen the construction of ecological and environmental protection science and technology innovation platforms, strengthen the construction of key laboratories, engineering technology centers, scientific observation and research stations and other science and technology innovation platforms, strengthen technology research and development promotion, and improve the scientific management of water

Guide enterprises to cooperate with scientific research institutions to promote the research and development of environmental protection technologies, the transfer and transformation of scientific and technological achievements, and their application.

Column 8 Key projects to enhance ecological environmental governance capabilities
Hunan Province Ecological Environment Monitoring Capacity Improvement Project. (1) Hunan Province Ecological Environment Monitoring Capacity Improvement Project (2020-2022). This includes the improvement of Hunan Province's ecological environment monitoring business capabilities and basic support construction. 2,077 sets of related instruments and equipment, 26 information platforms, and 618 station buildings. 35 monitoring vehicles were replaced, 22,600 m2 of monitoring buildings were newly built, and 8 monitoring buildings were renovated. (2) Monitoring capacity of county-level monitoring facilities in Hunan Province Power construction project. Ecological Sky Eye Project. (1) Carry out the construction of Hunan's natural ecological big data collection system, establish a terrestrial ecosystem monitoring network, and build a "sky-ground" integrated three-dimensional perception Internet of Things with Beidou, drones, lidar, infrared monitoring, smart emergency Internet of Things, ground monitoring stations and other natural ecological big data systems as the main body. (2) Build an ecological protection red line supervision platform for the whole province to grasp the interference of human activities in the province's natural reserves and ecological protection red lines in real time, and regularly evaluate and assess the ecological environment status, ecosystem structure, ecological function, ecological sensitivity, resource and environmental carrying capacity, and ecological protection and restoration effects. Law enforcement standardization construction project. Equipped in accordance with the requirements of the "Guiding Standards for the Standardization of Comprehensive Administrative Law Enforcement Equipment for Ecological and Environmental Protection (2020 Edition)" There are five major categories of standardized configurations, including law enforcement vehicles, personal mobile law enforcement equipment, on-site law enforcement auxiliary equipment, communication and office equipment, and information equipment. Talent team capacity improvement project. Regularly carry out ecological civilization ideological education for party and government leaders at all levels, and conduct governance capacity training and skill improvement education for leaders of municipal bureaus and municipal sub-bureaus of the ecological environment system. Ecological environment information construction project. Build a "big support platform", "big data center" and "big application" for ecological environment protection that is jointly built and shared by the whole province and interconnected. Use the system to achieve "one chess game, one integration" in the province's information capacity building. Ecological environment science and technology support project: Implement major ecological environment scientific research projects and promote technology and scientific research construction.

8. Planning and implementation of key projects

Focus on key planning areas, key tasks and measures, and implement key projects in eight directions, including "clear water", "blue sky", "clean land", agriculture and rural areas, heavy metals, ecological protection and restoration, risk prevention, and capacity building.

Plan and reserve key projects. Guided by the eight key projects, refine the project composition according to the key task categories in each direction, establish the "Hunan Province's 14th Five-Year Plan Ecological and Environmental Protection Key Project Reserve" (hereinafter referred to as the Project Reserve) to promote the project implementation of planning.

Strengthen project management. In accordance with the principles of "highlighting key points, performance-oriented, closed-loop management, and standardized and efficient", based on planning goals and key tasks, strengthen the evaluation and screening of key projects, establish a screening and dynamic adjustment mechanism for projects to be included in the database, and adjust the project reserve library in a timely manner.

For projects in the reserve pool, we will give priority to arranging relevant special funds, advance key projects in steps, and promote the implementation of planned tasks.

Strengthen expenditure responsibility. Establish a project management mechanism with provincial planning, municipal and prefectural implementation, and joint funding, and raise funds for engineering projects through multiple channels. Rationally divide expenditure rights and implement expenditure responsibilities of local governments at all levels. Improve the diversified ecological environment investment mechanism and actively guide social capital to participate.

We will actively innovate various investment and financing methods and vigorously promote the marketization of ecological and environmental governance.

IX. Measures to ensure implementation of the plan

Strengthen organizational leadership. Establish a division of labor and cooperation mechanism for provincial departments to promote the implementation of the plan, clarify the division of responsibilities, strengthen the organizational leadership of the implementation of the plan, strengthen the guidance, coordination and supervision functions, and ensure the smooth implementation of the plan. Governments at all levels should be responsible for the ecological environment quality of their jurisdictions.

The objectives and main tasks determined in this plan shall be combined with local realities to formulate and implement the "14th Five-Year" ecological and environmental protection plans of various cities, prefectures, counties, and districts, decompose and implement the planning objectives and tasks, and ensure the smooth realization of the planning objectives. When various regions and departments prepare relevant plans, they must be well coordinated with this plan.

Prepare special plans or action plans for various elements and fields of the ecological environment and implement the goals and tasks.

Increase investment. According to the "Implementation Plan for the Reform of the Division of Financial Powers and Expenditure Responsibilities between the Province and Cities and Counties in the Field of Ecological Environment in Hunan Province", the financial powers and expenditure responsibilities below the municipal level in the field of ecological environment shall be reasonably divided, the provincial power in cross-regional ecological and environmental protection shall be appropriately strengthened, and the city and county governments shall be strengthened.

The ability to guarantee basic public services for ecological and environmental protection. Governments at all levels should make ecological and environmental protection a key area of fiscal expenditure and establish and improve a normalized and stable mechanism for local fiscal investment in ecological and environmental governance. Actively promote cooperation between the government and social capital to attract social capital to participate in ecological and environmental protection.

Protection items.

Strengthen publicity and guidance. Actively carry out publicity of ecological and cultural policies and regulations, experience and results, use the new media matrix of provincial, municipal and county levels, innovate expression methods and communication methods, enrich ecological and cultural works, and do a good job in typical reports. Actively expand the social influence of June 5th Environment Day and deeply promote

We will launch a series of themed practice activities, including "Beautiful China, I am an actor" and "Green Guards Go to Hunan", increase the number of units with environmental protection facilities open to the public, accelerate the creation of ecological and environmental education bases, and strengthen the cultivation and guidance of social environmental protection organizations.

Conduct planning evaluation. The Provincial Department of Ecology and Environment will work with relevant departments to coordinate the progress of the objectives, key tasks, and key projects of this plan. At the end of 2023 and 2025, a mid-term evaluation and summary evaluation of the implementation of this plan will be conducted respectively, and the evaluation results will be reported to the Provincial People's Government.

The results will be reported to the government and announced to the public.

Appendix: Key Projects in the Hunan Province's 14th Five-Year Plan for Ecological and Environmental Protection

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ÿ Notice of the General Office of the People's Government of Hunan Province on the issuance of the "Hunan Province's 14th Five-Year Plan for Ecological and Environmental Protection".doc

ÿ Key projects of the Hunan Province's 14th Five-Year Plan for Ecological and Environmental Protection.doc

Related interpretation

"Hunan Province's 14th Five-Year Plan for Ecological and Environmental Protection" was released on 2021-10-24

Hunan Provincial Department of Ecology and Environment interprets the "Hunan Province's 14th Five-Year Plan for Ecological and Environmental Protection" 2021-10-25

Policy Brief: Blue skies, clear waters, and clean land! The blueprint for Hunan's ecological and environmental protection in the next five years is here 2021-10-27

Video 1 "Hunan Province's "14th Five-Year Plan" Ecological Environmental Protection Plan" is released. By 2025, the province's green and low-carbon development level will be significantly improved 2021-10-25

Organizer: General Office of the People's Government of Hunan Province

Organizer: Hunan Provincial Government Service and Big Data Center

Co-organizer: Hunan Daily

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